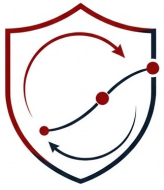
Principles of Robot Autonomy I

Planning and Control: Trajectory optimization, direct and shooting methods, LQR

Stanford University



SIA Lab
Safe and Intelligent Autonomy Lab

IPRL

Principles of Robot Autonomy I

Announcements:

- Lab section 2 this week
- HW 1 due this Thurs, Oct 8
- Slides uploaded to canvas before lecture
- Total enrolled: 160, total waitlist: 50

Lecture 5:

- PRM*, RRT* and re-wiring
- Trajectory optimization
- Direct, indirect, and shooting methods
- LQR, iLQR

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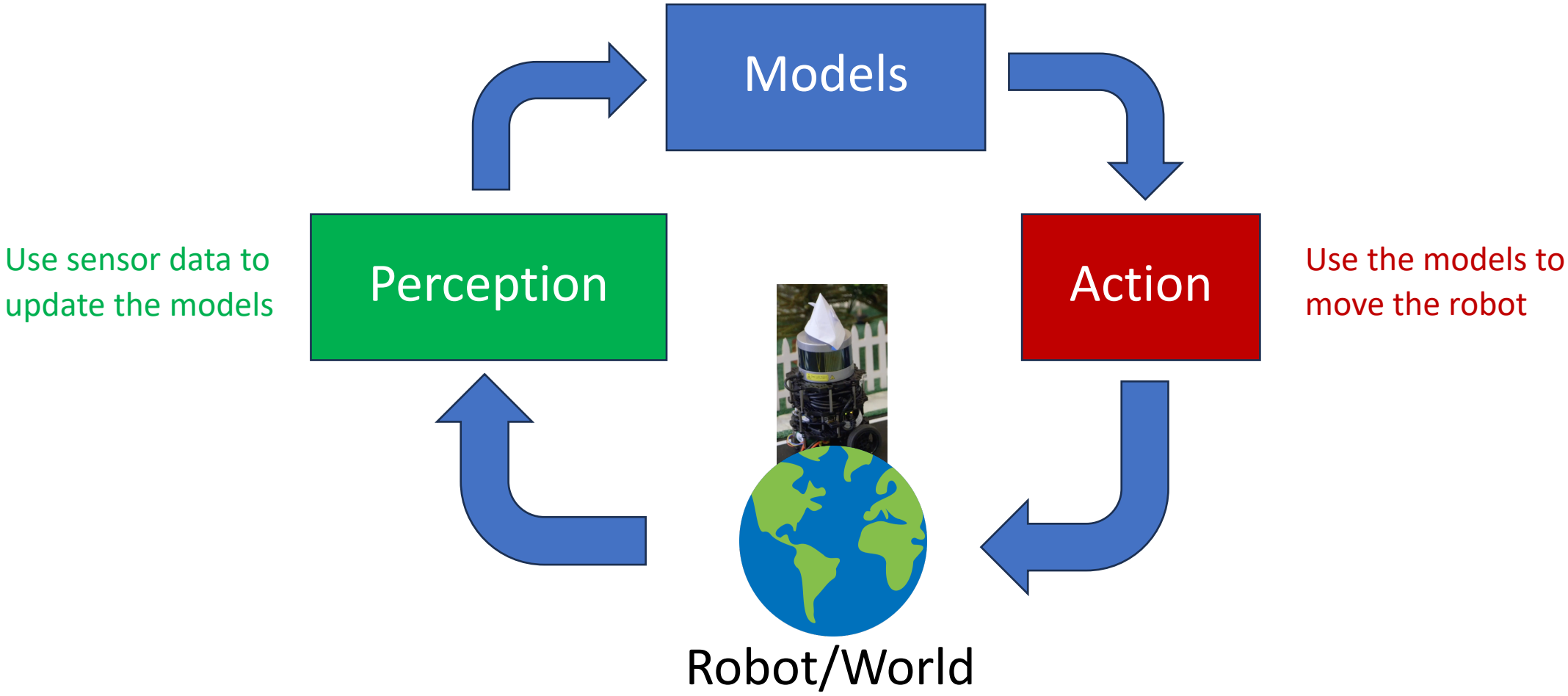
SIA Lab
Safe and Intelligent Autonomy Lab

MSL

IPRL

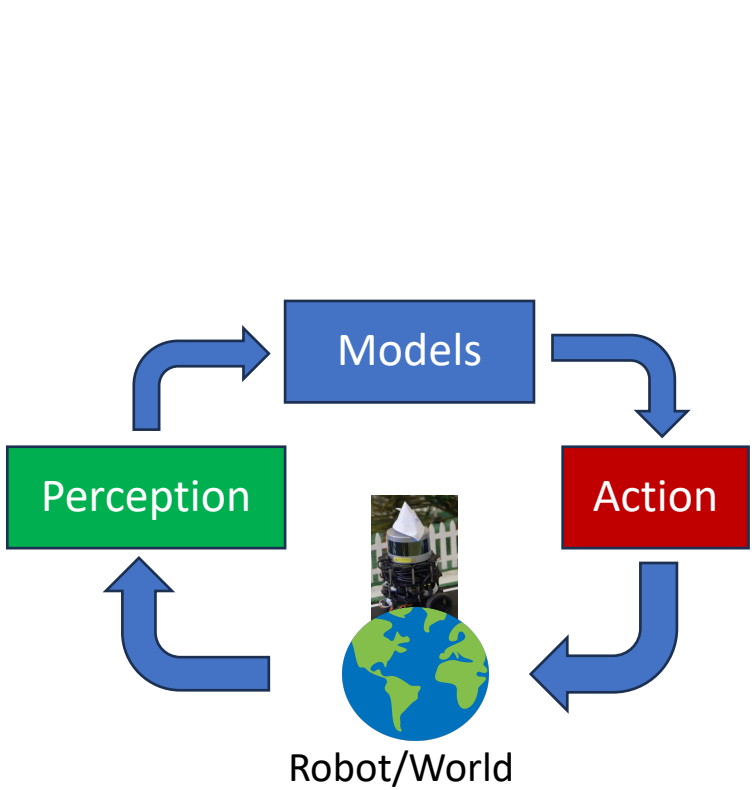
ASL
Autonomous Systems Lab

Full Stack Autonomy: the Perception-Action Loop



Action: Planning and Control Stack

Use the models to move the robot



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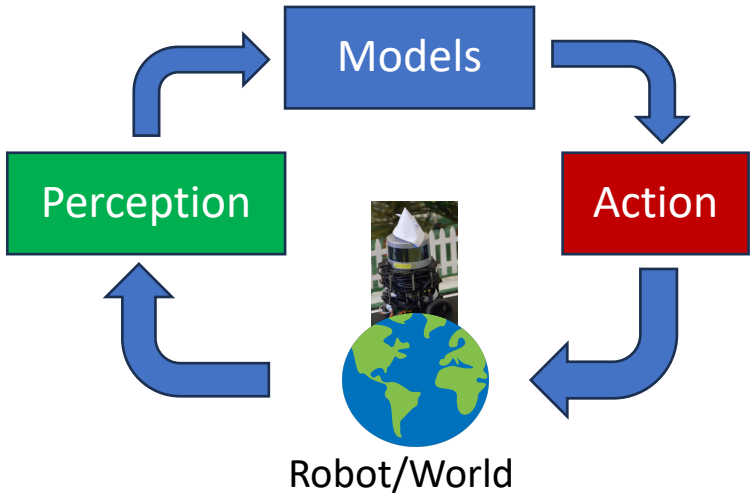

Path planning:

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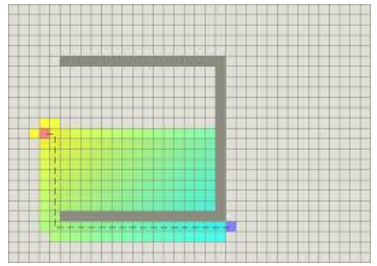

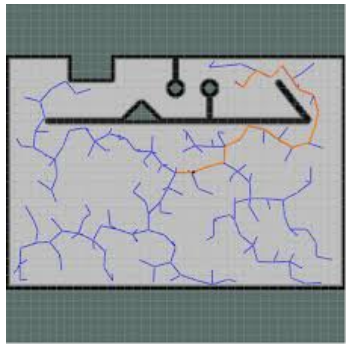

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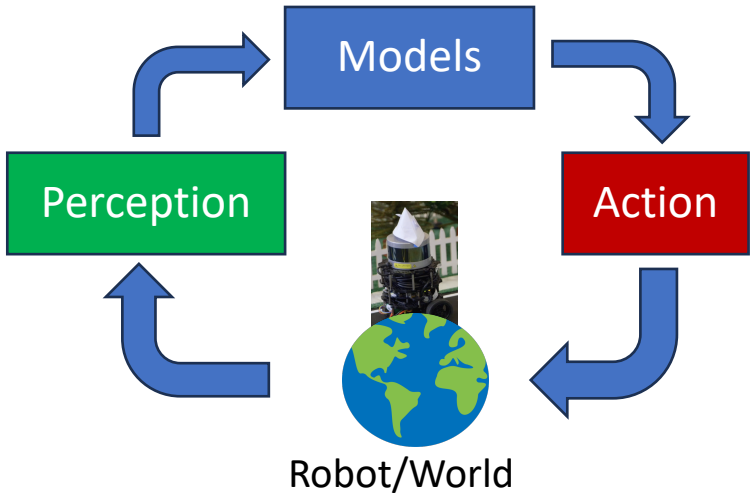

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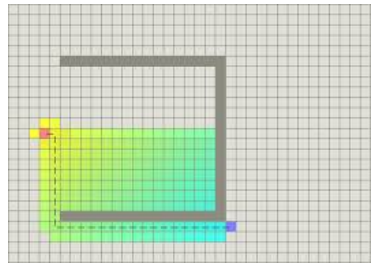

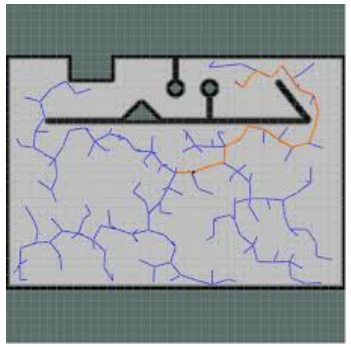

Trajectory optimization:

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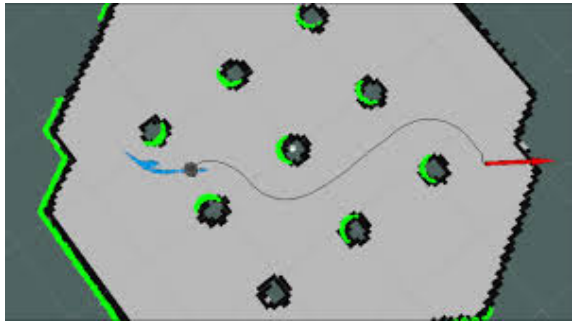

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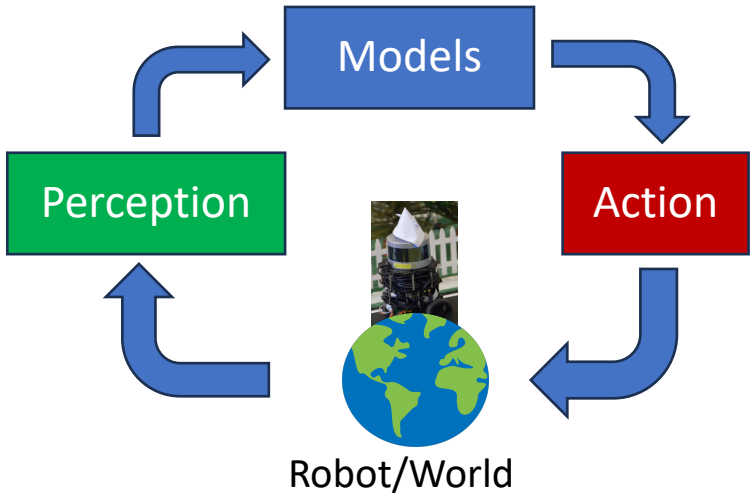

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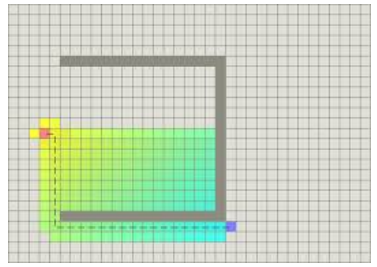

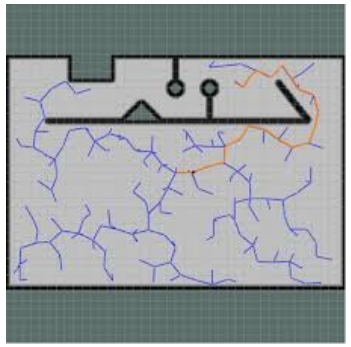

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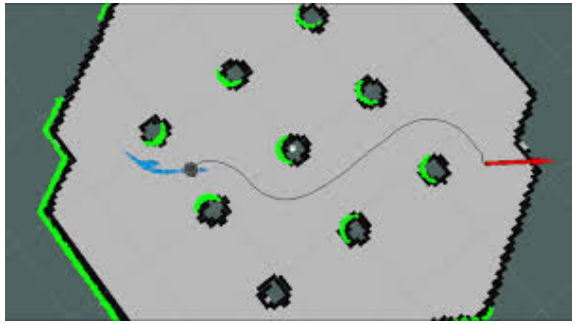

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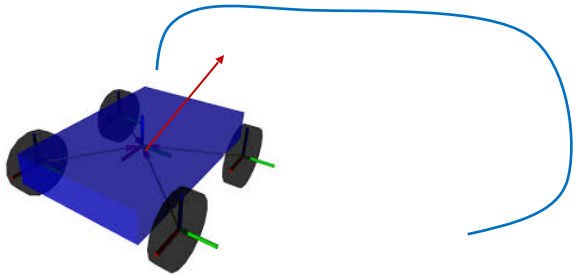

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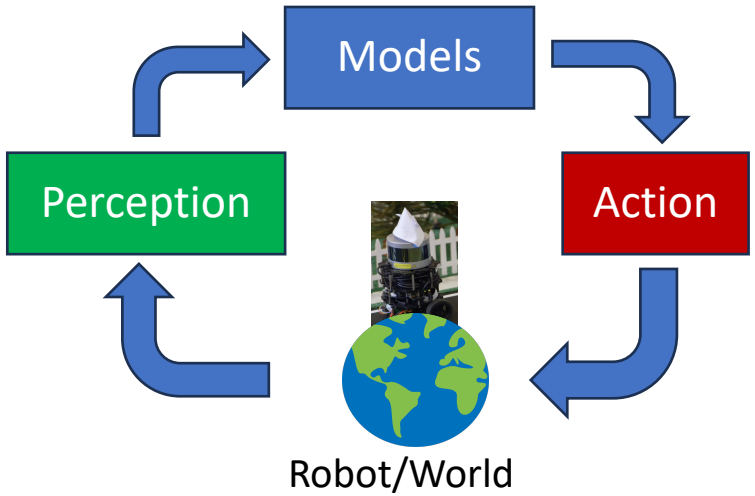

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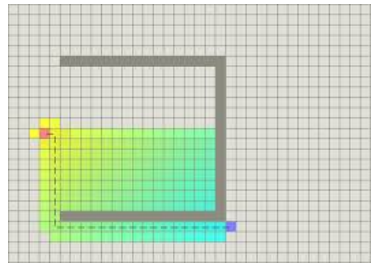

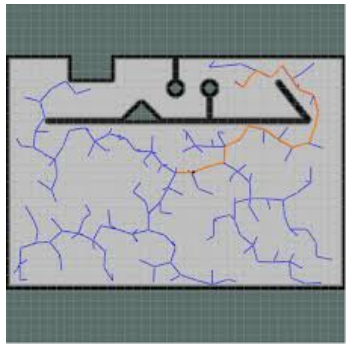

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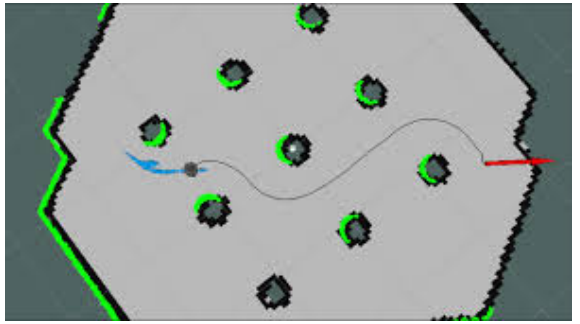

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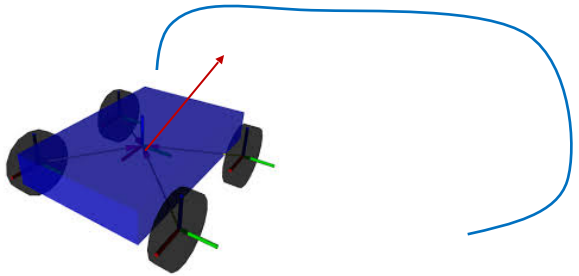

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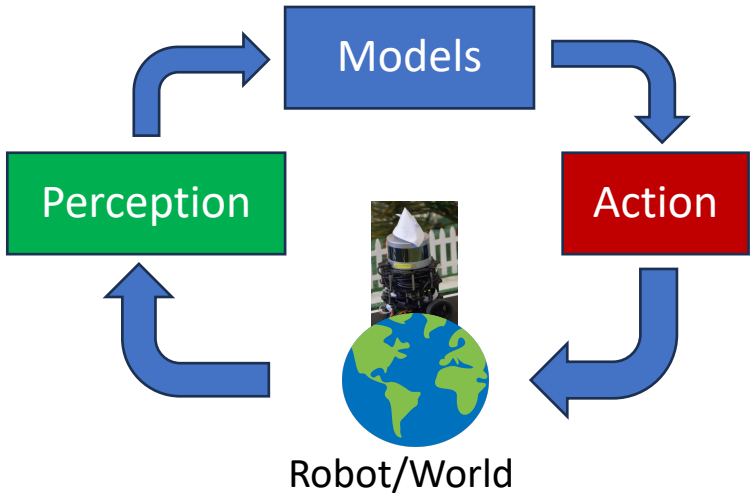

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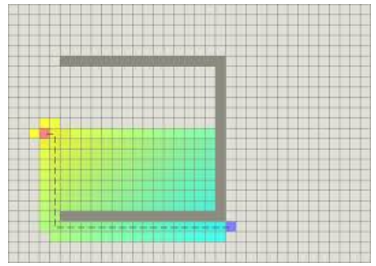

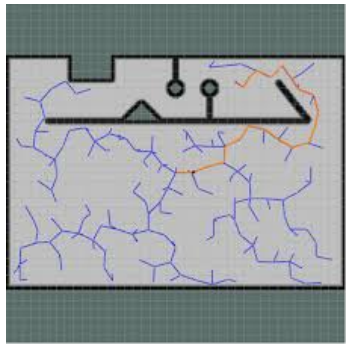

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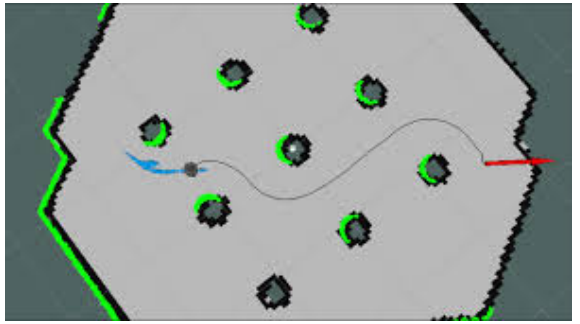

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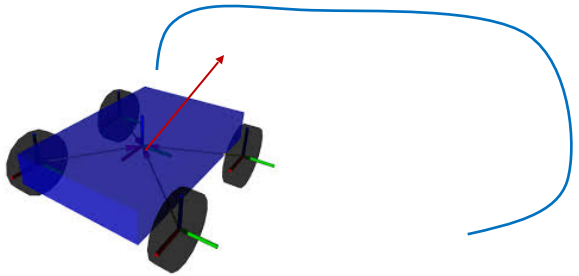

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Map representation

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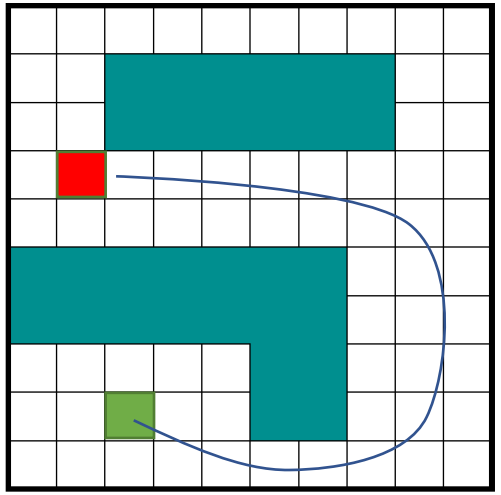
Graph search planning, A*



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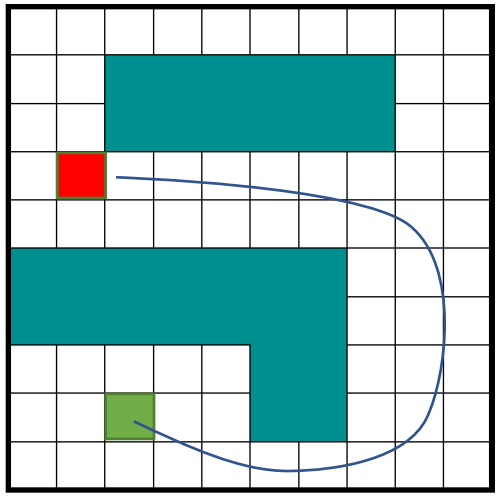
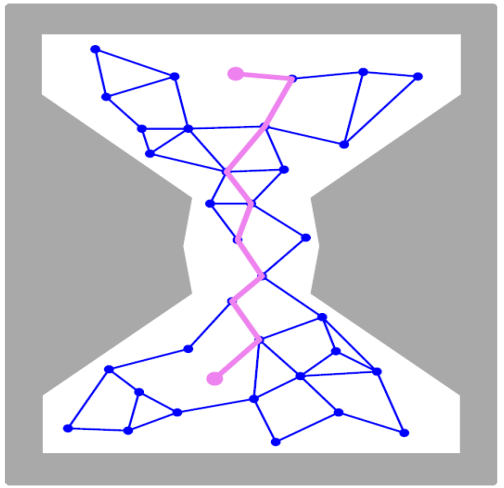


Planning on random graph, RRT*, PRM



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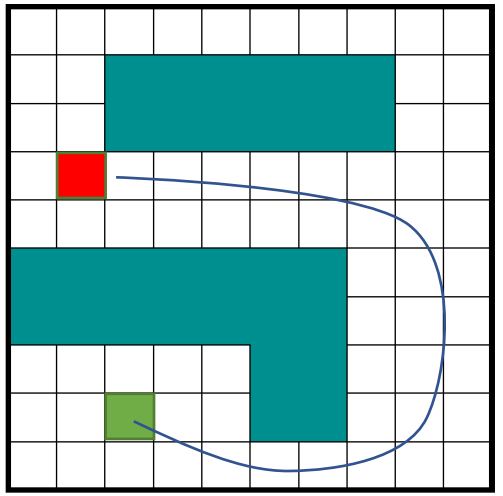

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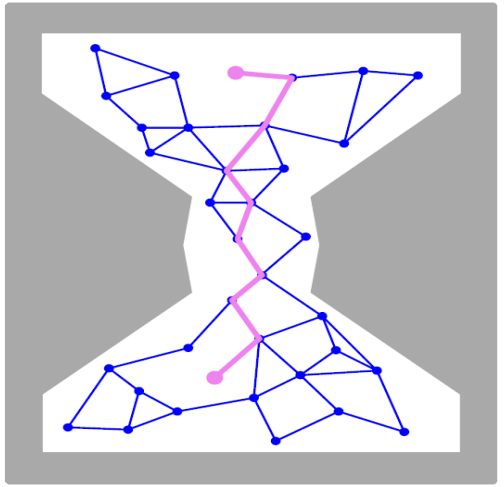

Goal: Find a path so the robot doesn't hit things!

Sampling-based planning

Challenges of explicit C-space computation motivated the development of sampling-based approaches

- Abandon the idea of explicitly characterizing C_{free} and C_{obs}
- Instead, capture the structure of C by random sampling
- Use a black-box component (collision checker) to determine which random configurations lie in C_{free}
- Use such a probing scheme to build a roadmap (a graph of possible paths) and then plan a path

Probabilistic roadmaps (PRM)

A multi-query planner, which generates a roadmap (**graph**) G, embedded in the free space

Preprocessing step:

1. Sample a collection of n configurations X_n; discard configurations leading to collisions
2. Draw an edge between each pair of samples $x, x' \in X_n$ such that $\|x - x'\| \leq r$
3. And straight-line path between x and x' is collision free

Given a query $s, t \in C_{free}$, connect them to G and find a path on the

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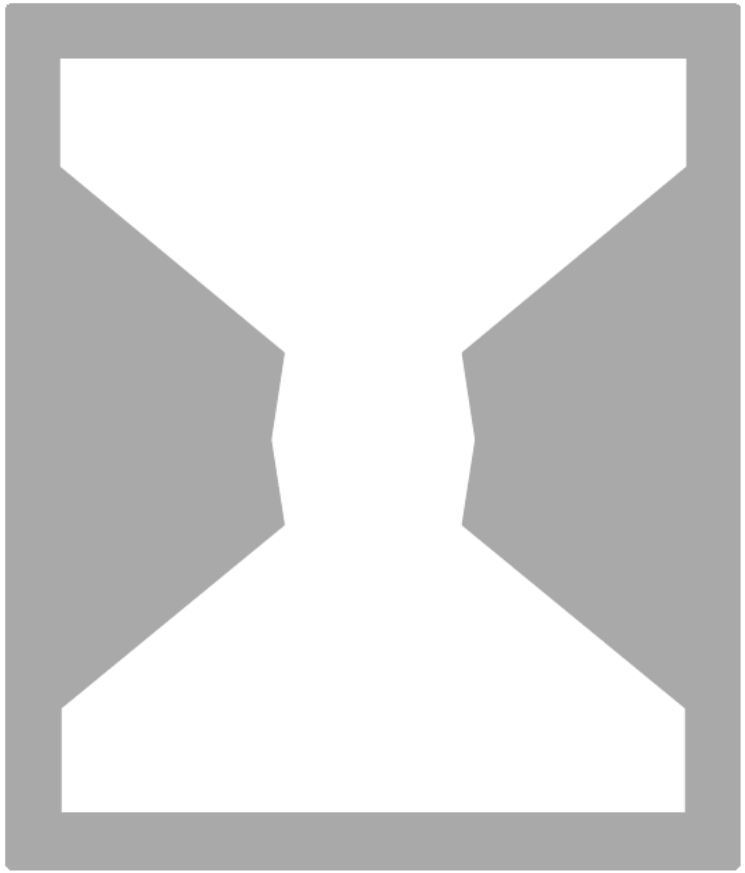

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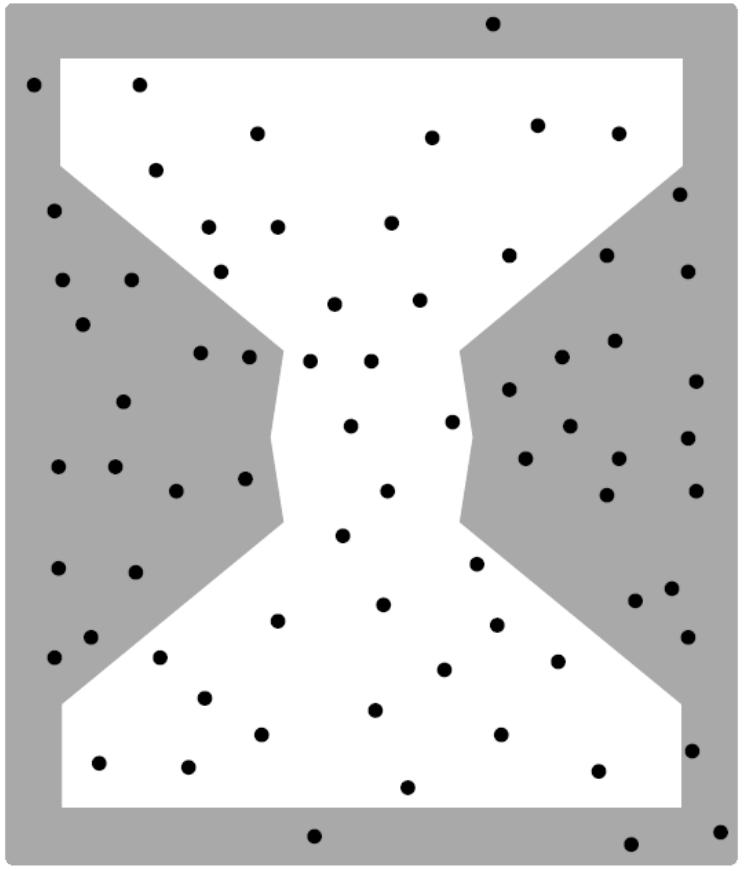

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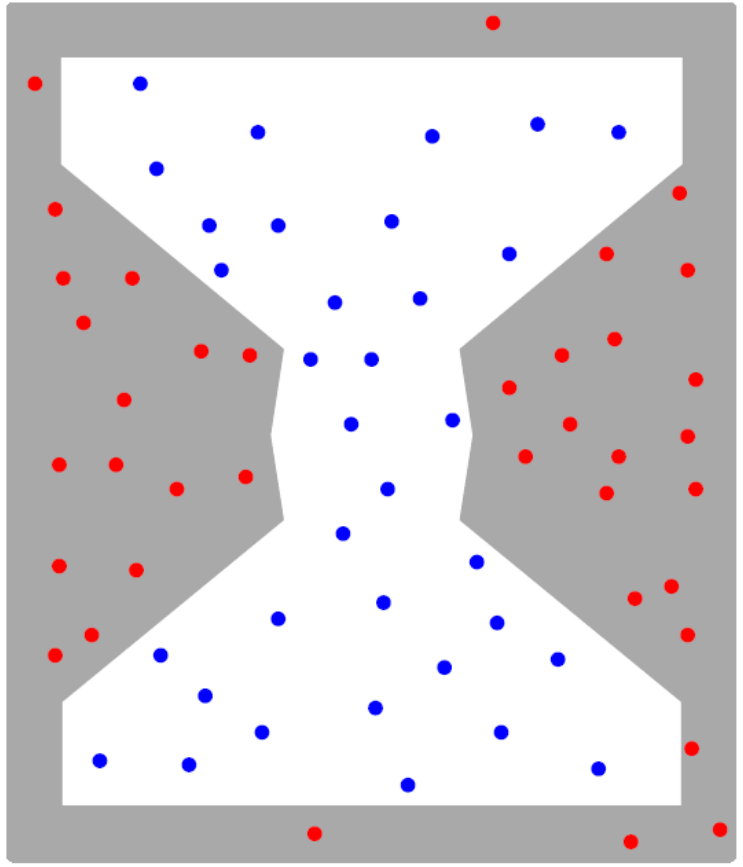

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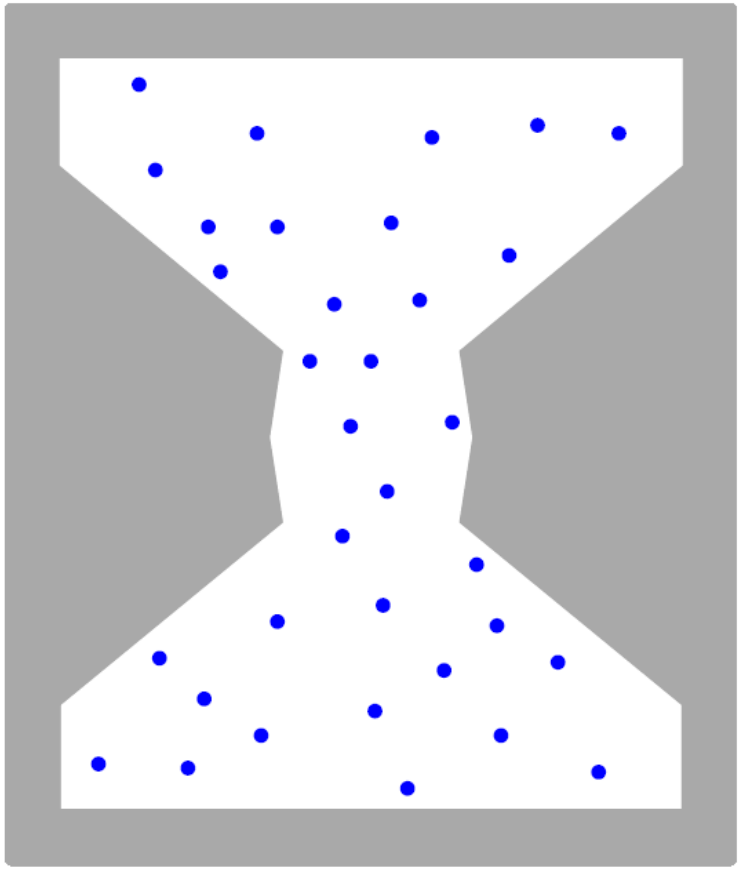

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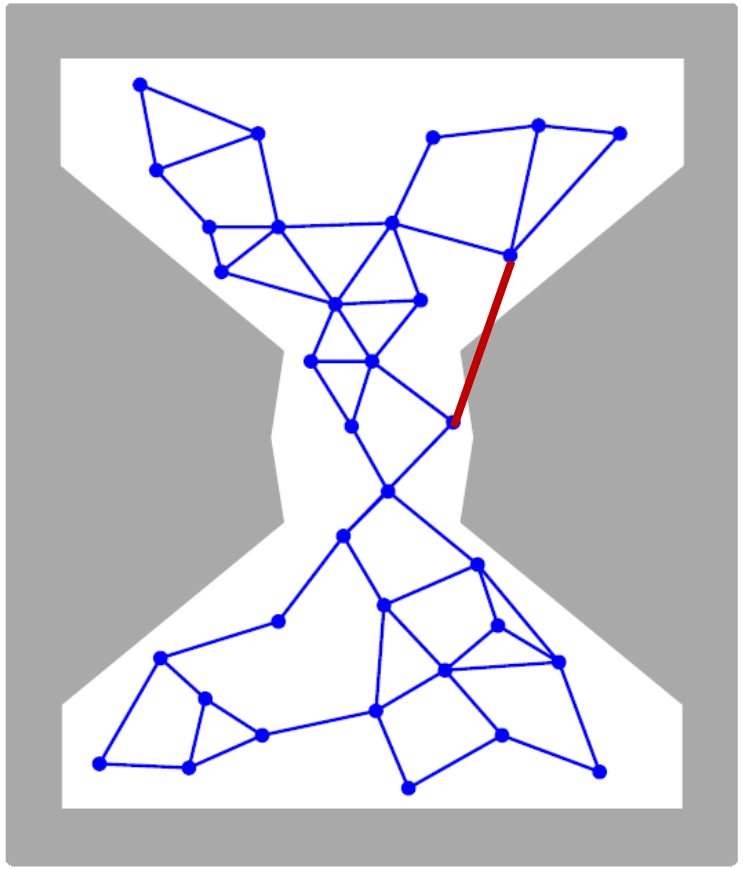

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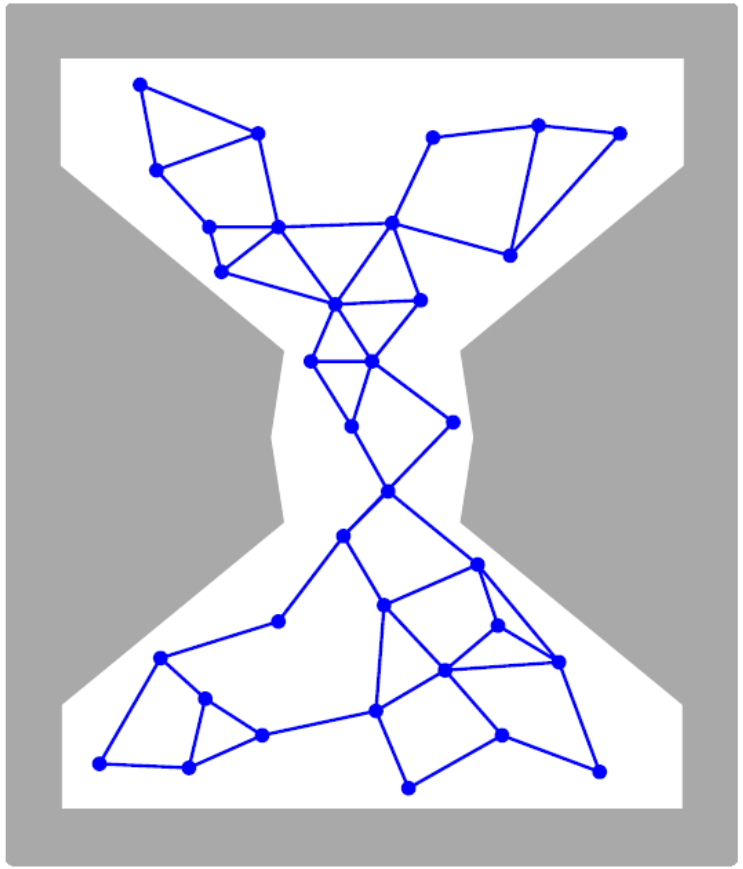

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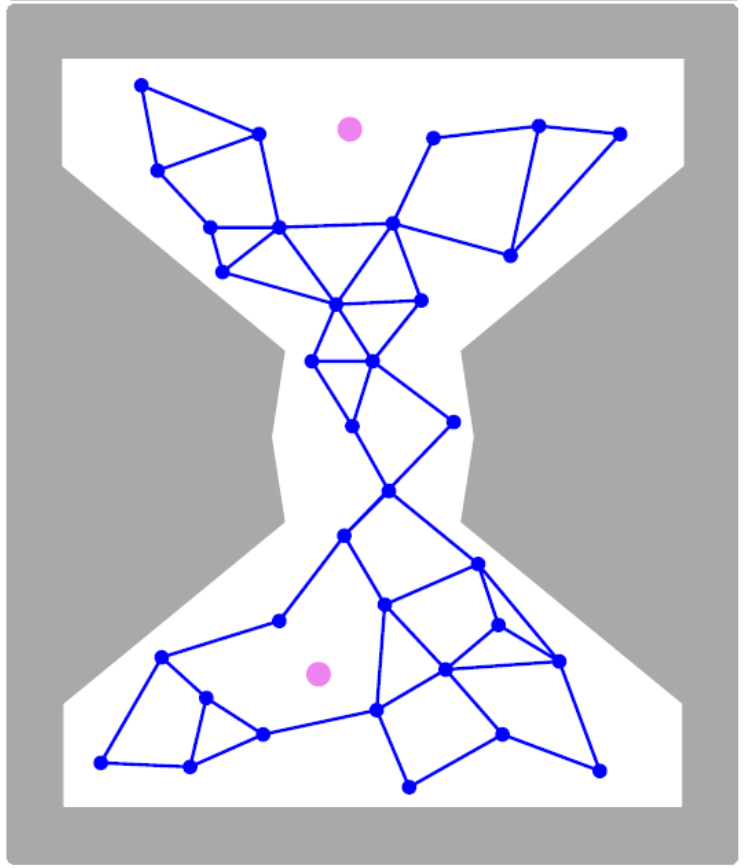

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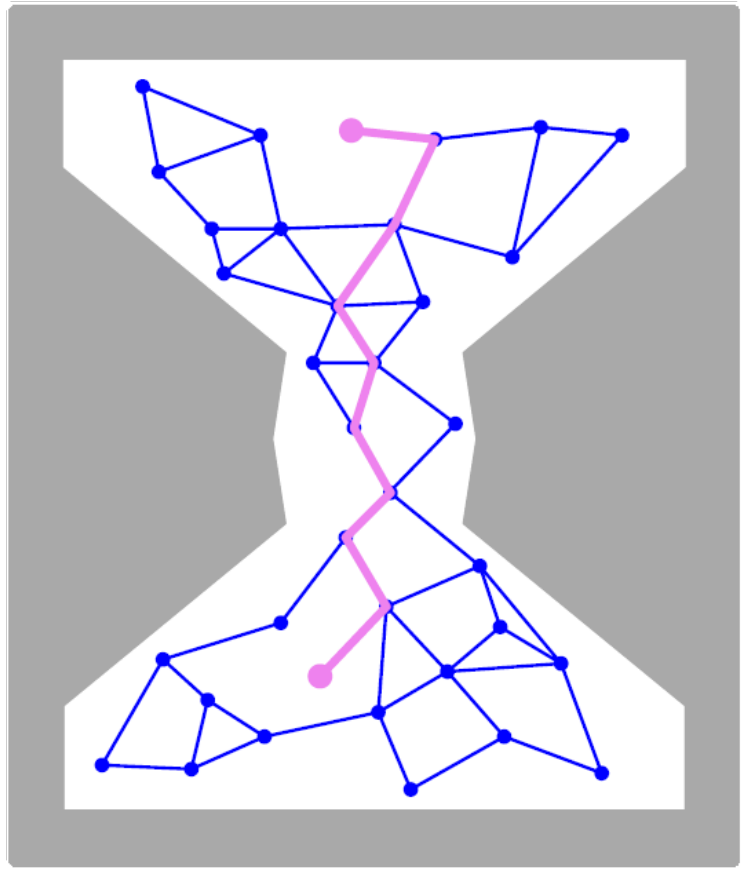

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Rapidly-exploring random trees (RRT)

A single-query planner, which grows a **tree** $T = (V, E)$, rooted at the start configuration s, embedded in C_{free}

Algorithm works in n iterations:

1. Sample configuration x_{rand}
2. Find nearest vertex x_{near} in T to x_{rand}
3. Generate configuration x_{new} in direction of x_{rand} from x_{near}, such that $\|x_{near} - x_{new}\| \leq \eta$ and $\{x \mid \alpha x_{near} + (1-\alpha)x_{new}, \alpha \in [0,1]\} \in C_{free}$
4. Update tree: $T = (V \cup x_{new}, E \cup (x_{near}, x_{new}))$

Occasionally, set x_{rand} to be the target vertex t; terminate when $x_{new} = t$

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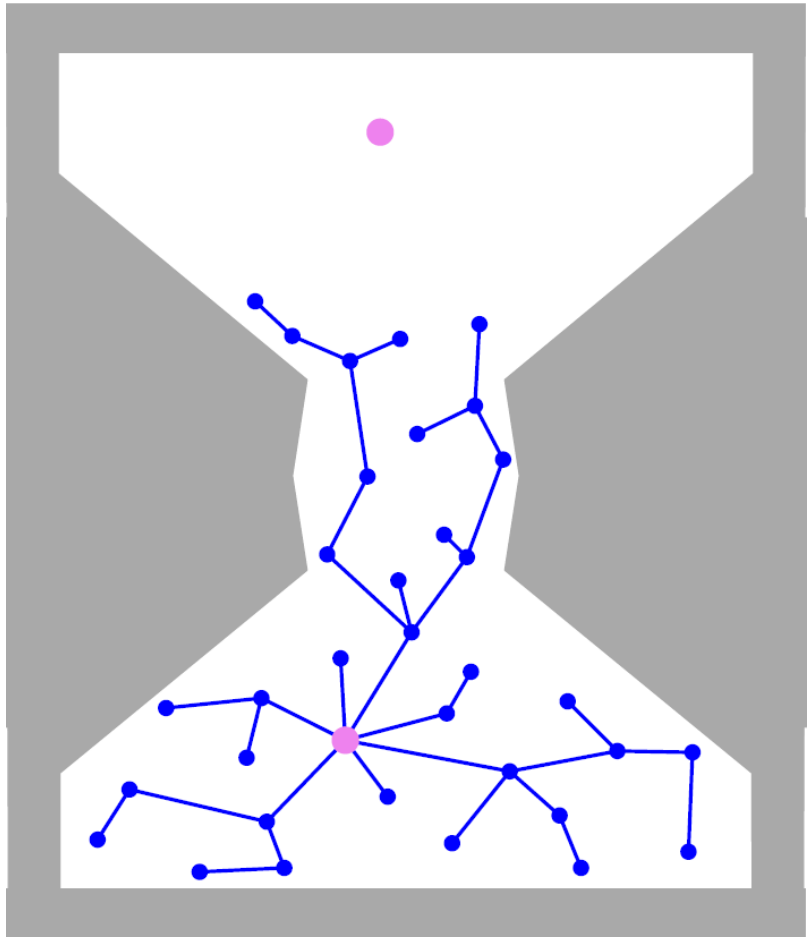

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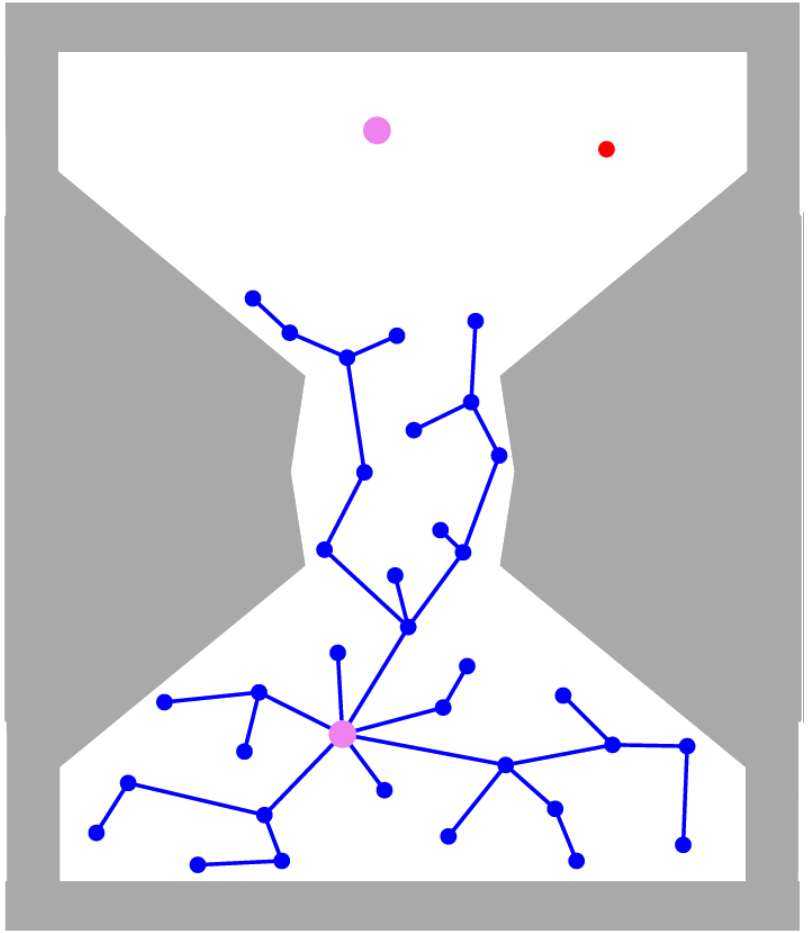

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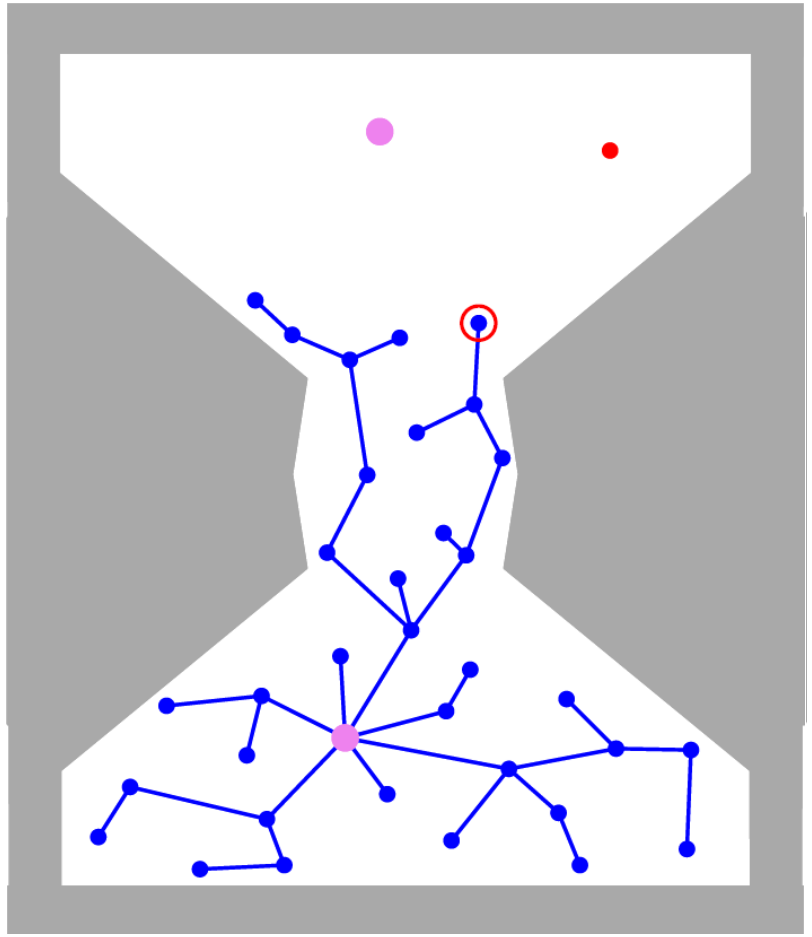

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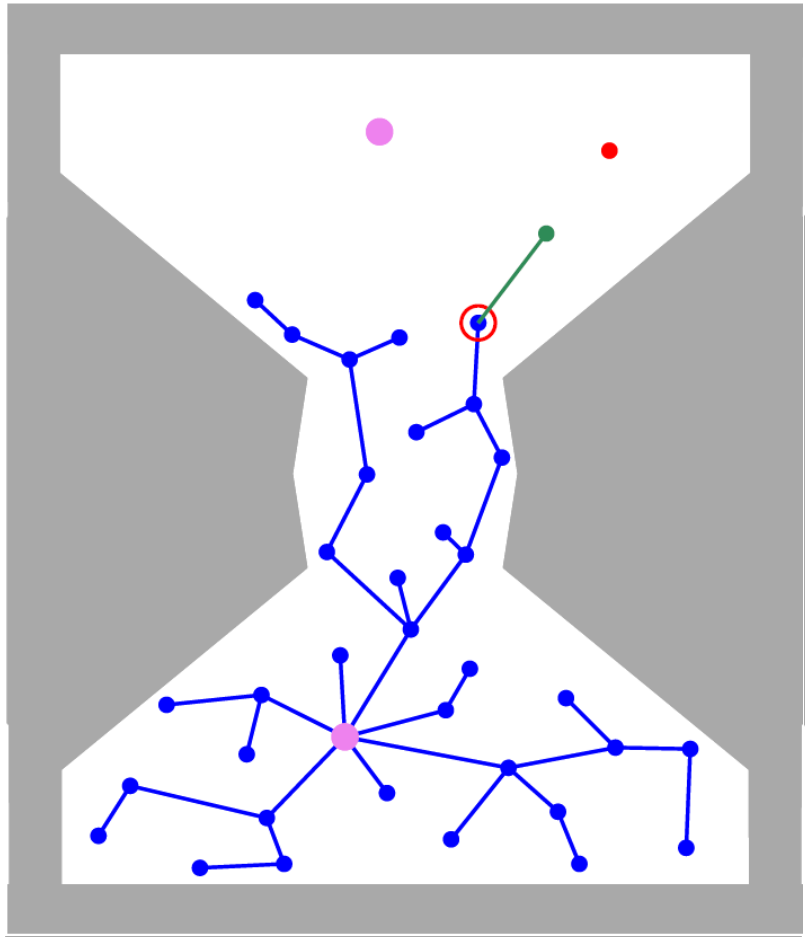

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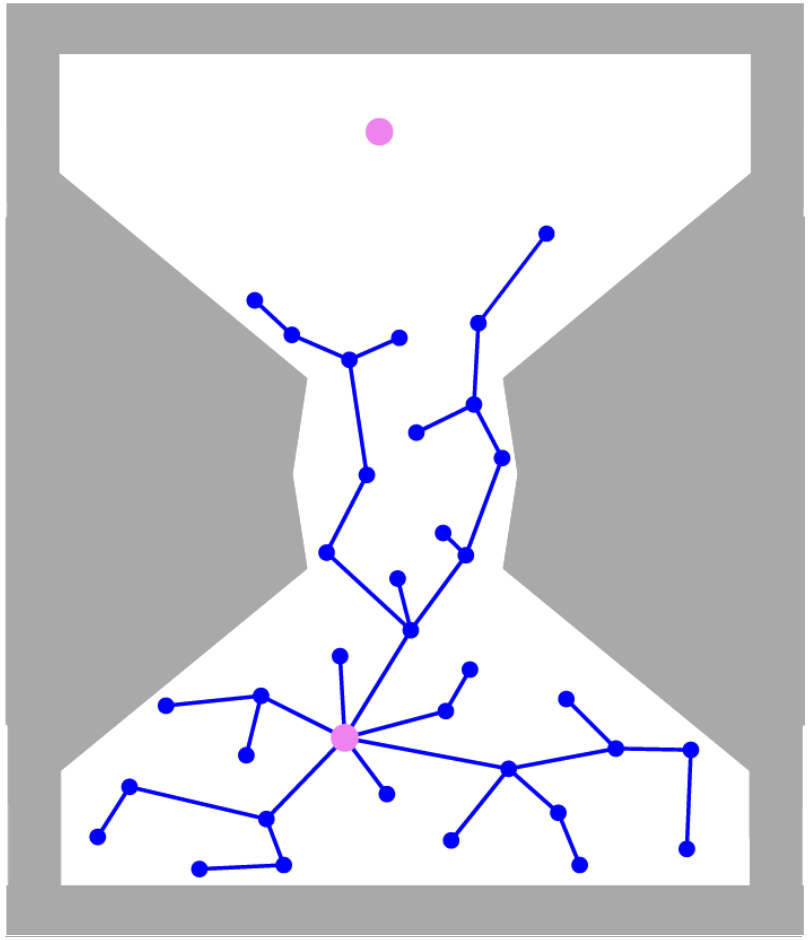

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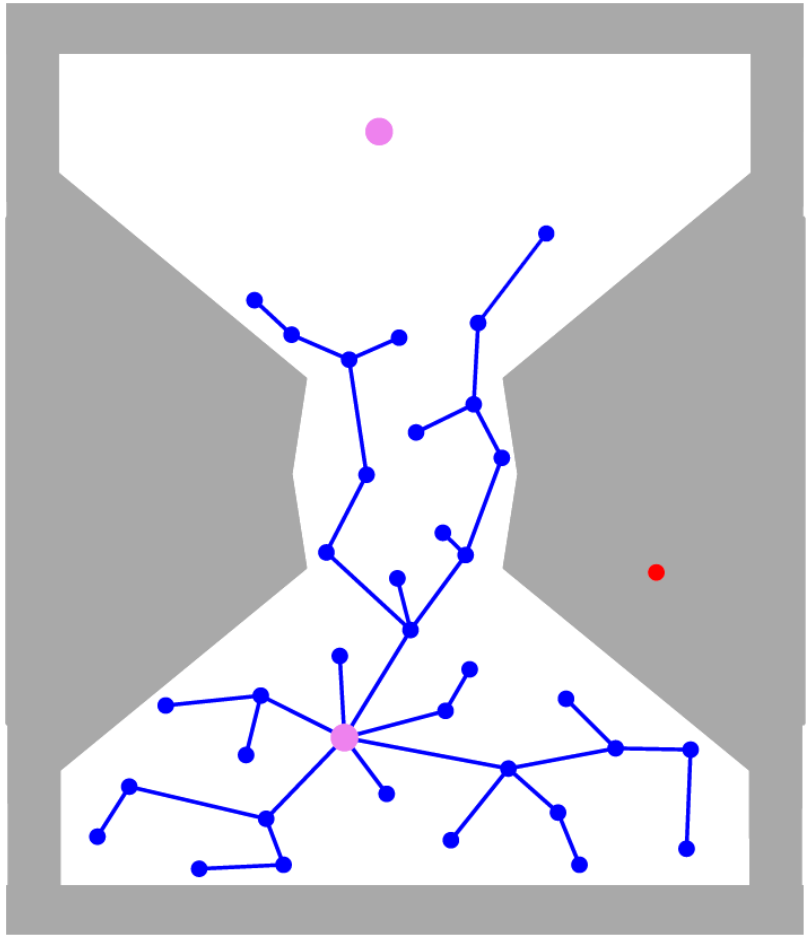

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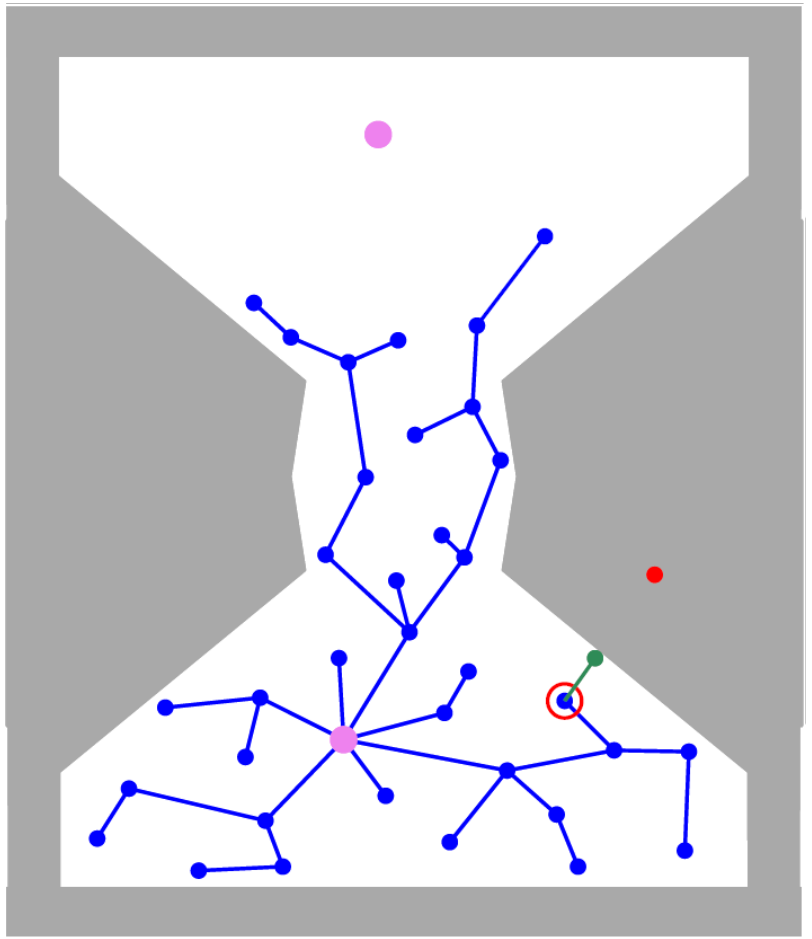

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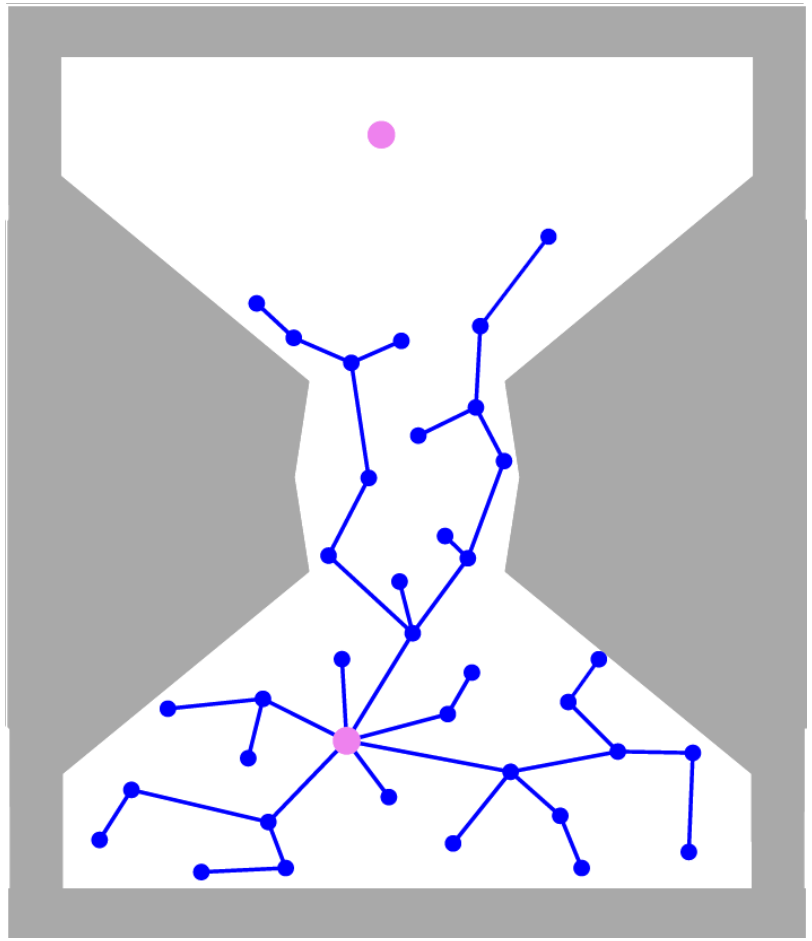

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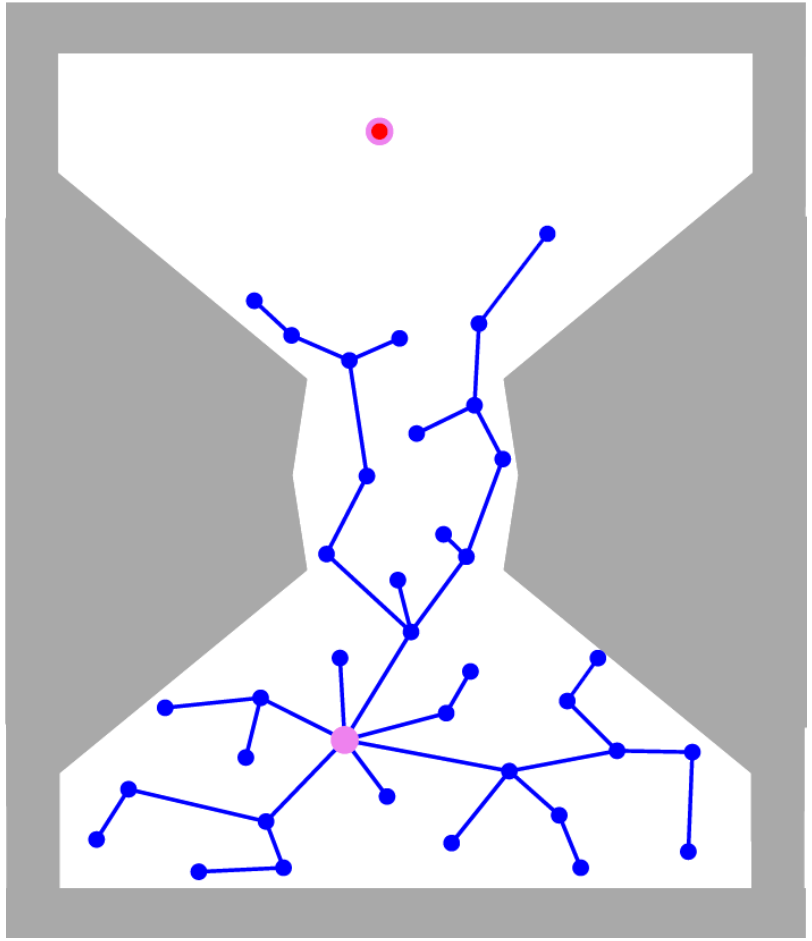

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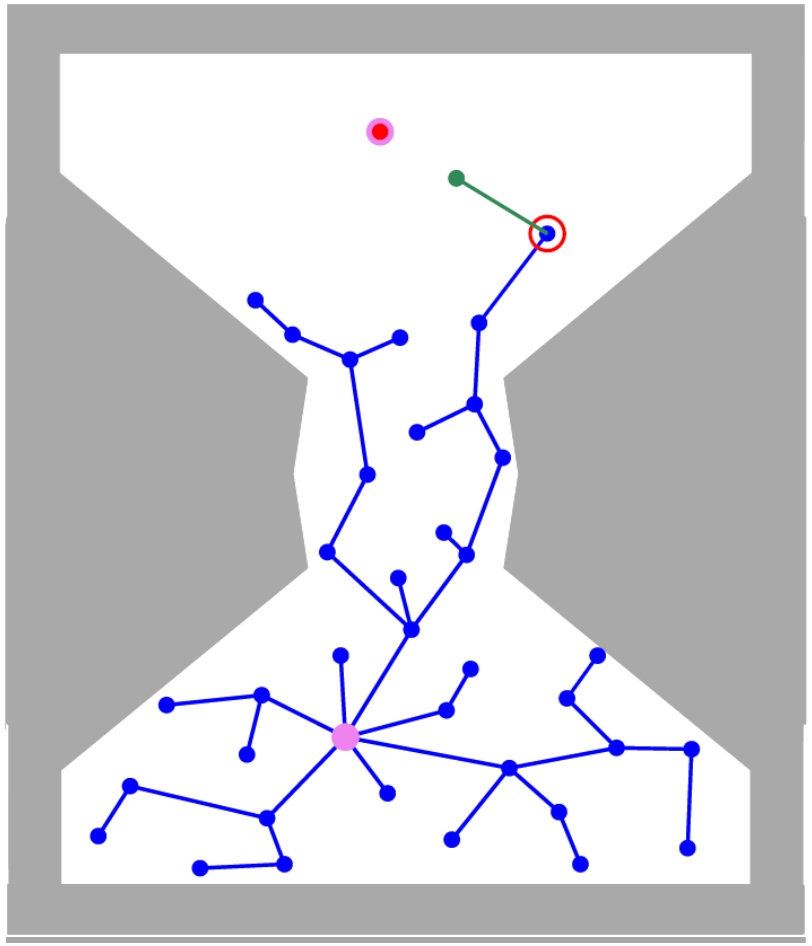

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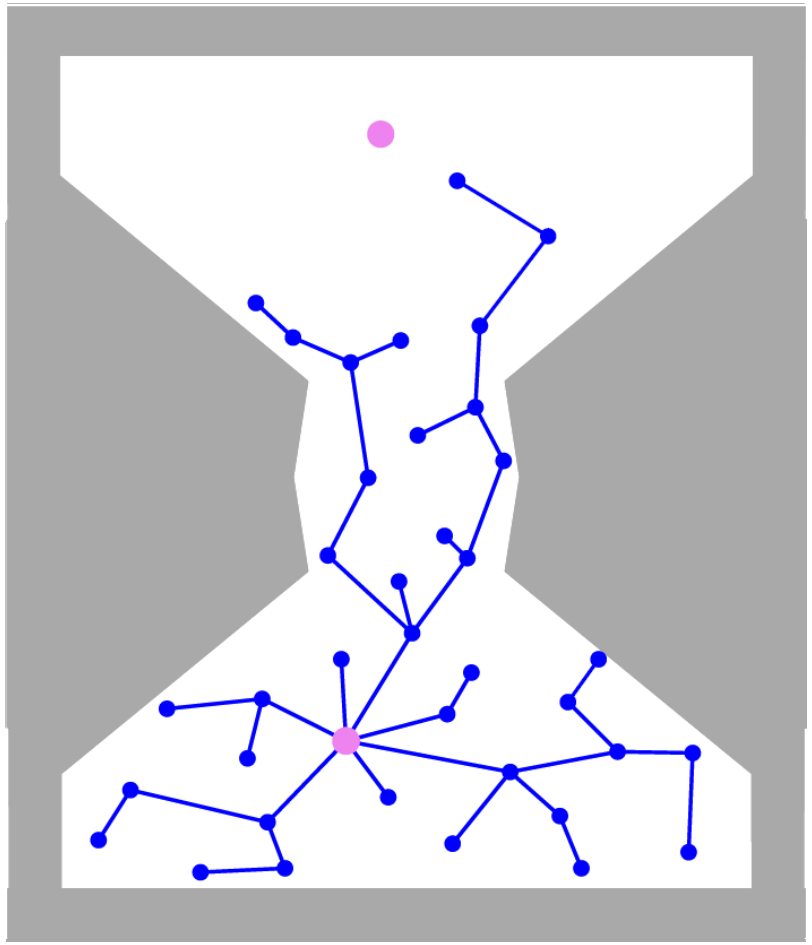


Theoretical guarantees: probabilistic completeness

Question: how large should the number of samples n be? We can say something about the asymptotic behavior:

Kavraki et al. 96: For PRM, with $r = \text{const}$, the probability of finding a path if one exists approaches 1 as $n \to \infty$.

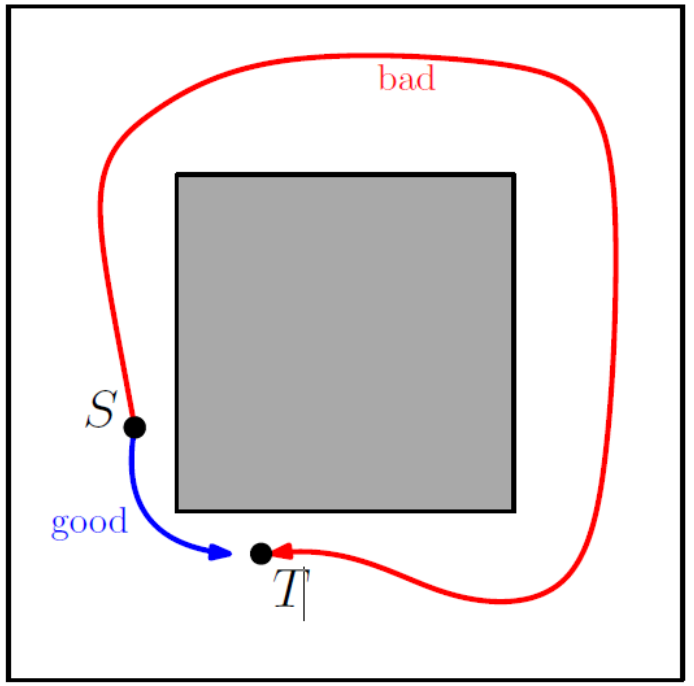
LaValle, 98; Kleinbort et al., 18: For RRT, the probability of finding a path if one exists approaches 1 as $n \to \infty$

* hypercube $[0,1]^d$, with $2 \leq d < \infty$

Theoretical guarantees: quality

Question: what can be said about the quality of the returned solution for PRM and RRT, in terms of length, energy, etc.?

- Nechushtan et al. (2011) and Karaman and Frazzoli (2011) proved that RRT can produce arbitrarily-bad paths with non-negligible probability: for example, RRT might prefer to take the long (red) way.
- This is because it does not **revise** connections with more efficient choices that might be found later (recall Dijkstra's, A*).

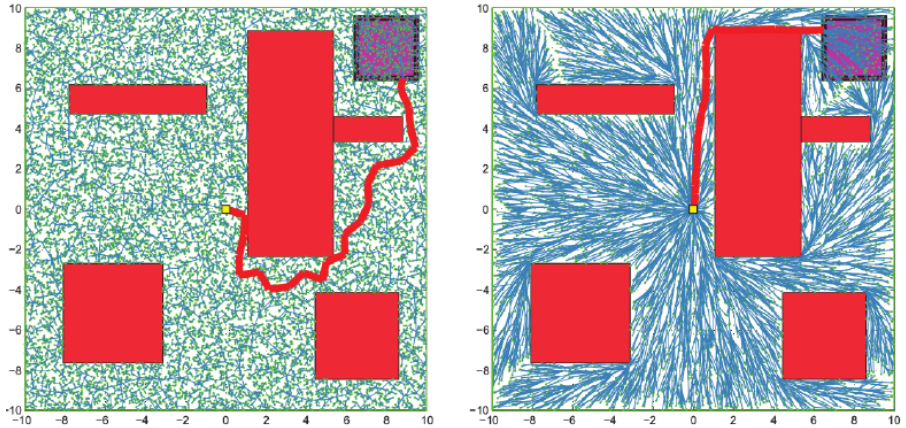


Theoretical guarantees: quality

Karaman and Frazzoli in 2011 provided the first rigorous study of optimality in sampling-based planners:

Theorem: The cost of the solution returned by PRM converges, as $n \to \infty$, to the optimum, when $r_n = \gamma\left(\frac{\log n}{n}\right)^{\frac{1}{d}}$, where γ only depends on d

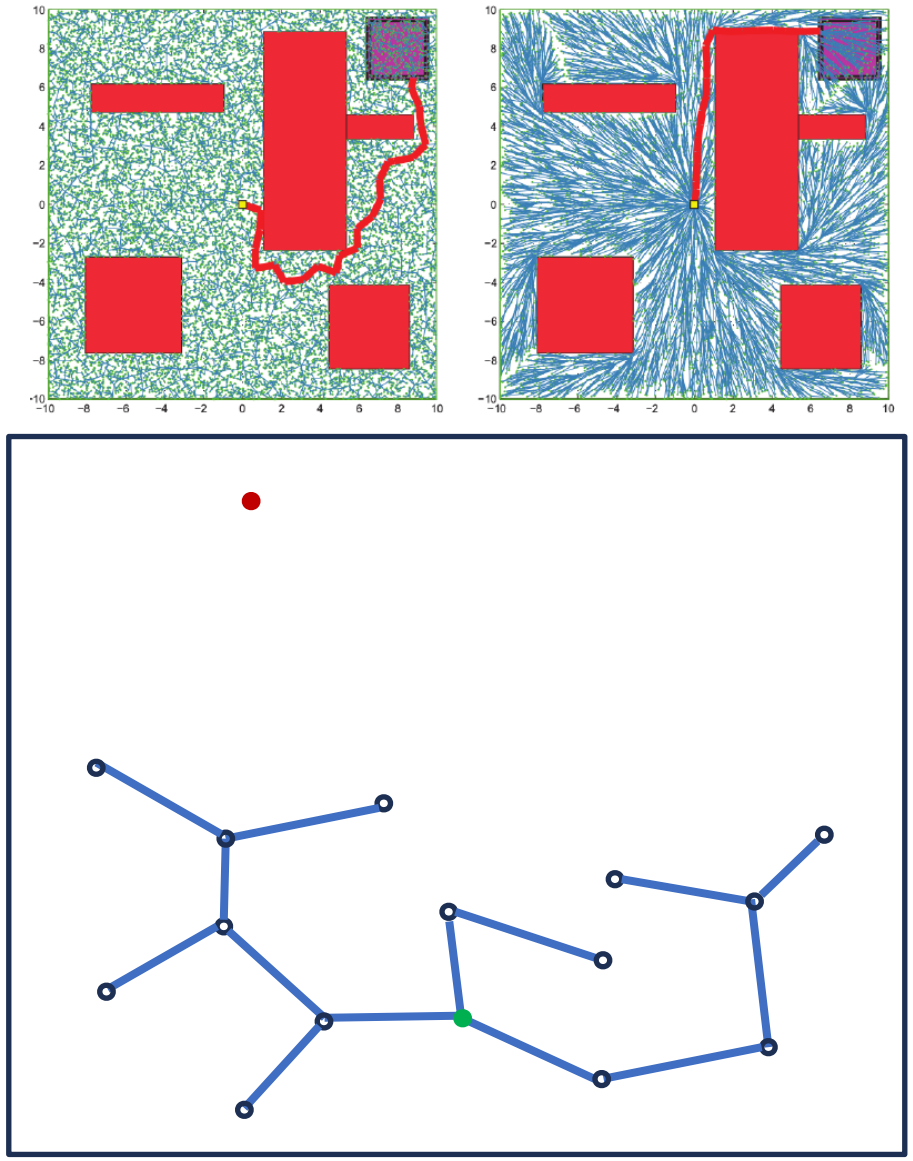
- KF11 also introduced an asymptotically optimal variant of RRT called RRT* (right)
- Introduced **re-wiring,** another name for label correcting, just like Dijkstra's/A*.



RRT* Re-wiring

- After connecting vertex x_{new} to the tree, check alternative connections to all potential parents in ball radius r_n
- If a lower cost path found through different parent, add edge to new parent, delete edge to old parent
- If a lower cost path found through x_{new} as parent, add new/delete old
- r_n shrinks as n grows, according to

$$r_n = \gamma \left(\frac{\log n}{n} \right)^{\frac{1}{d}}$$



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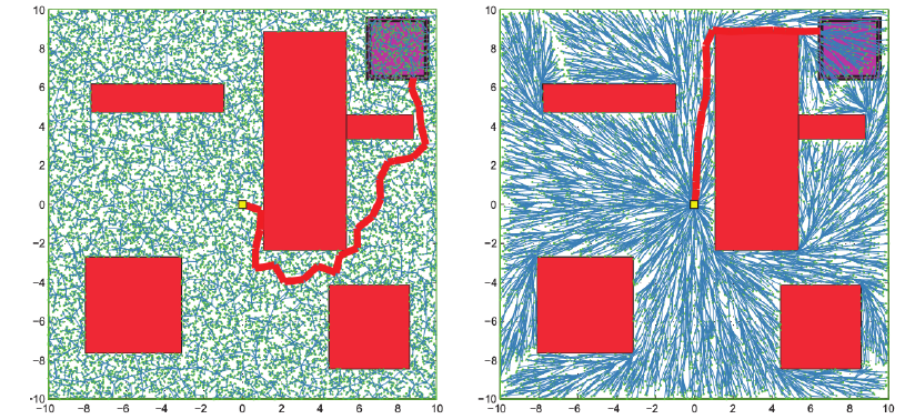

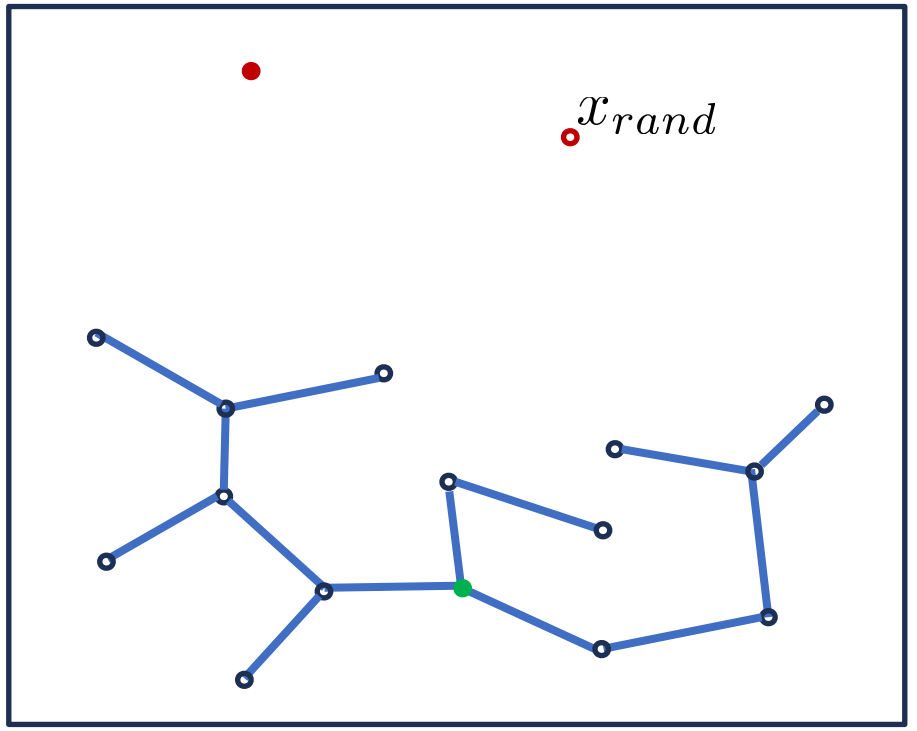

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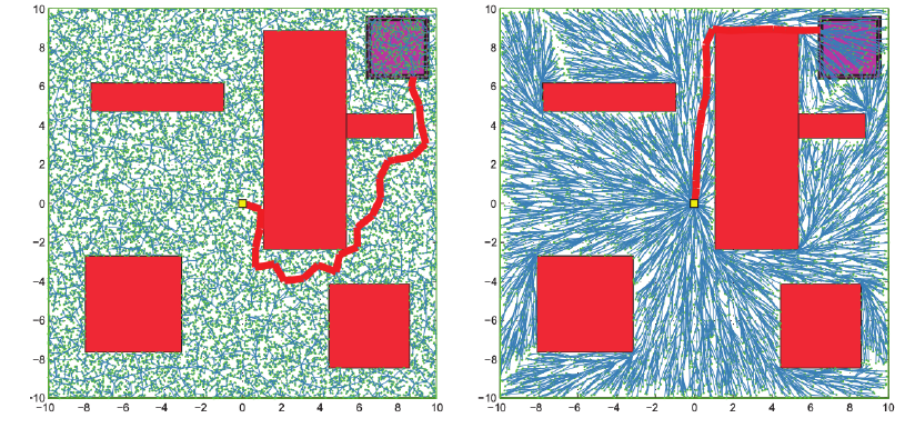

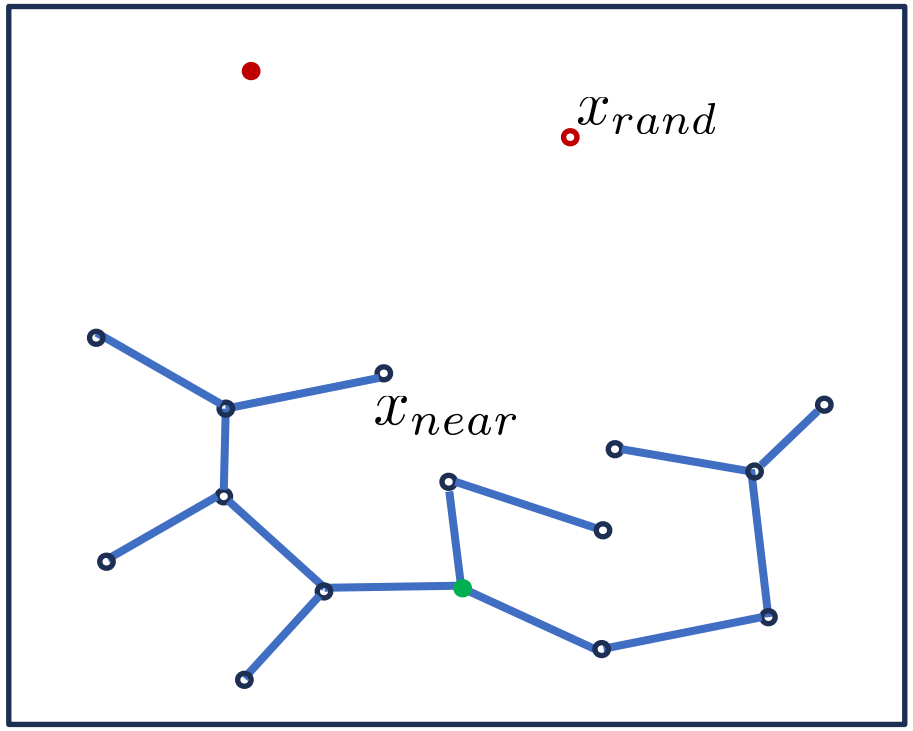

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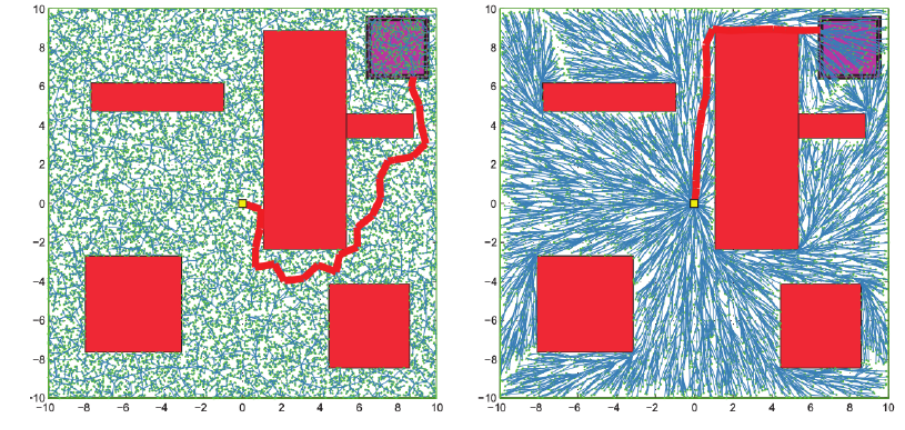

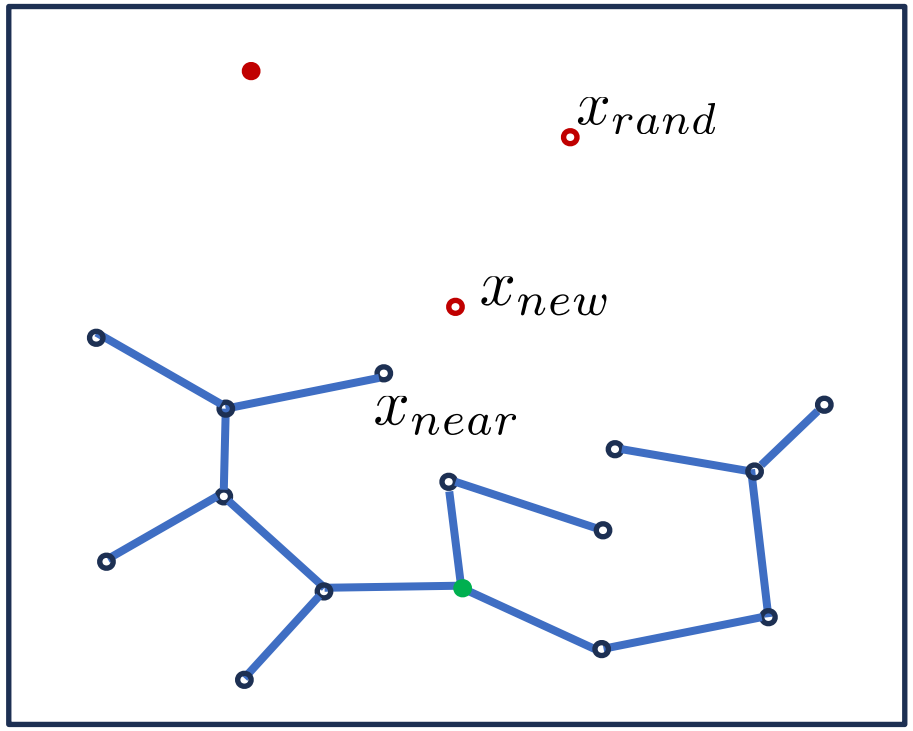

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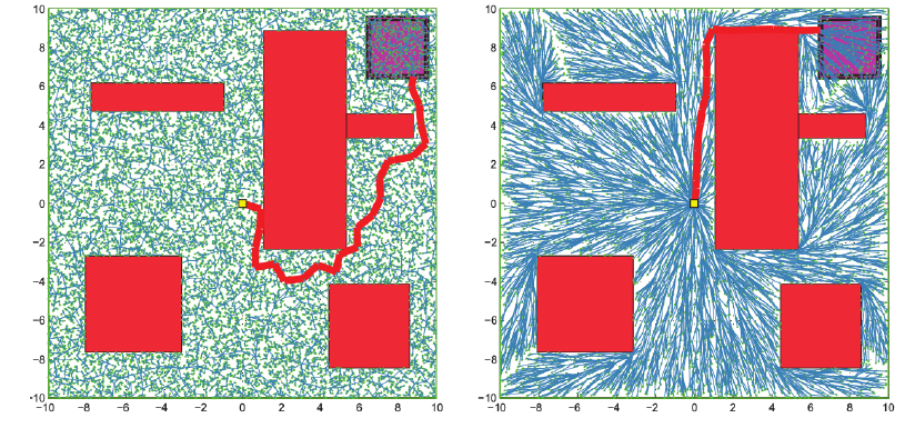

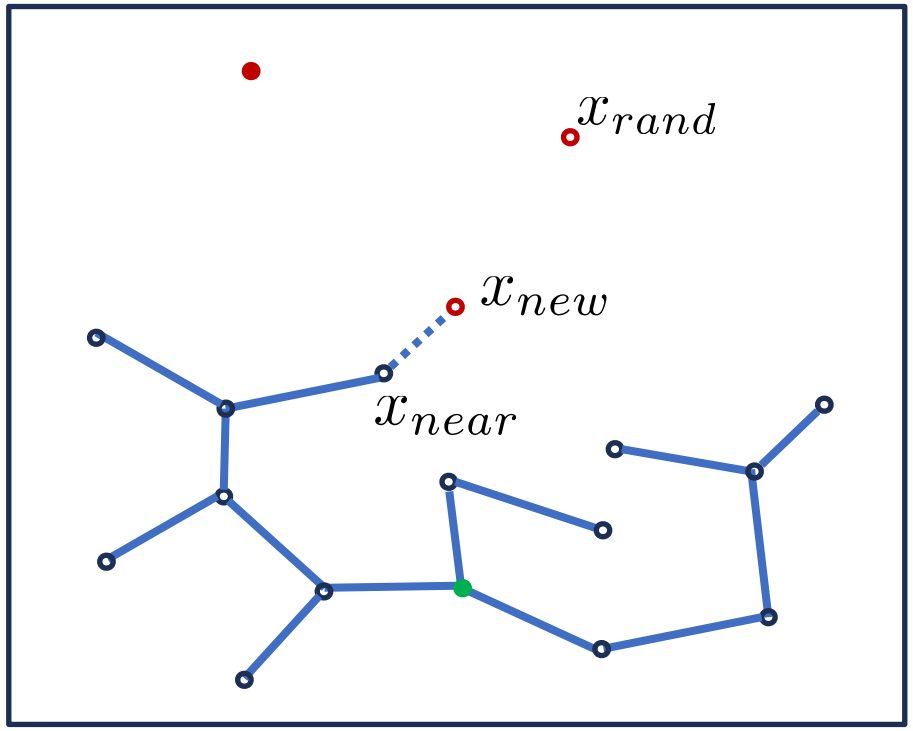

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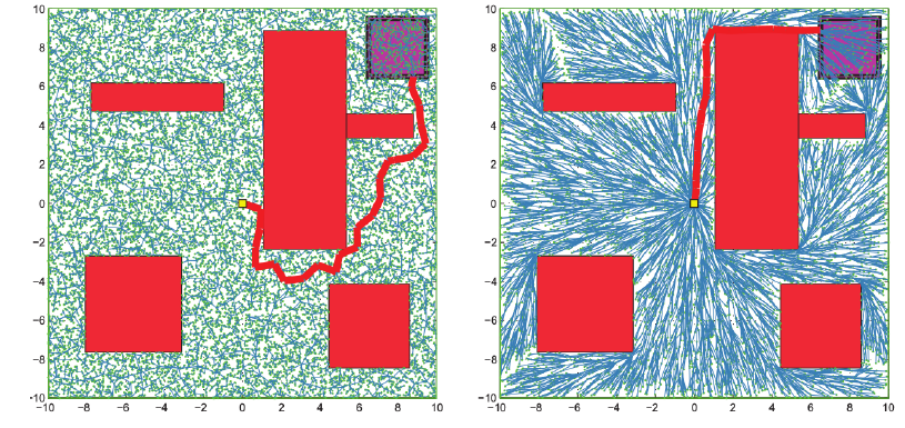

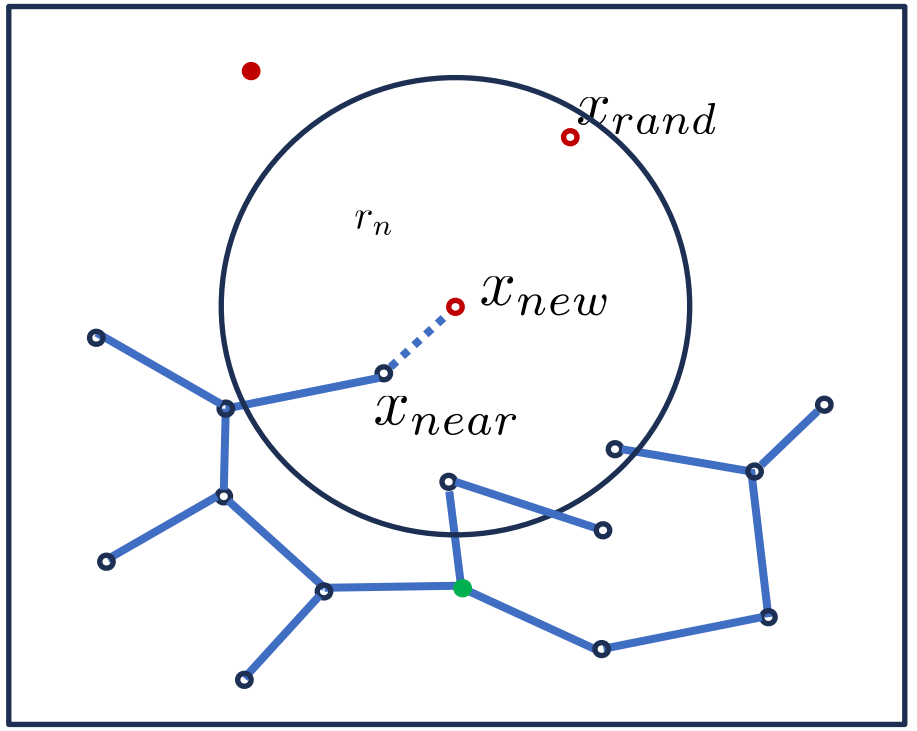

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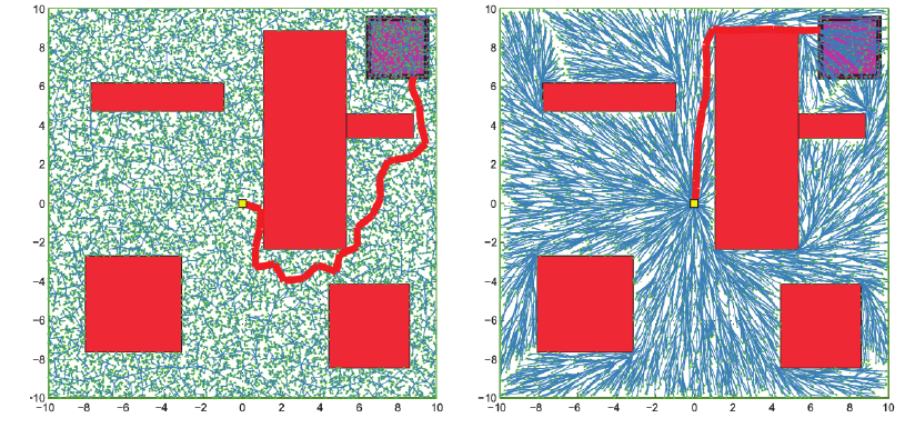

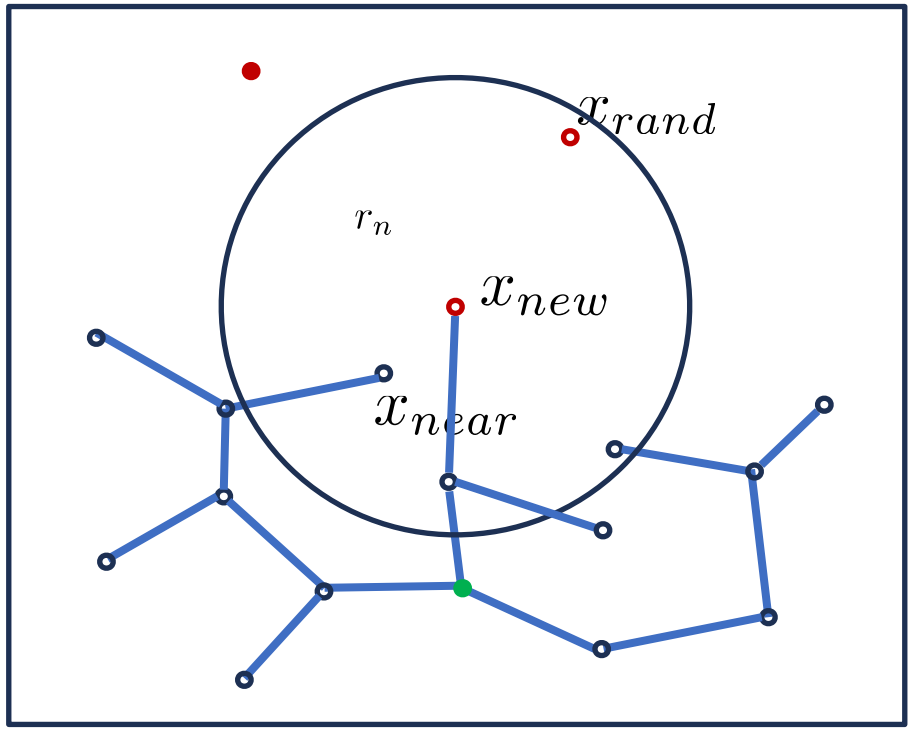

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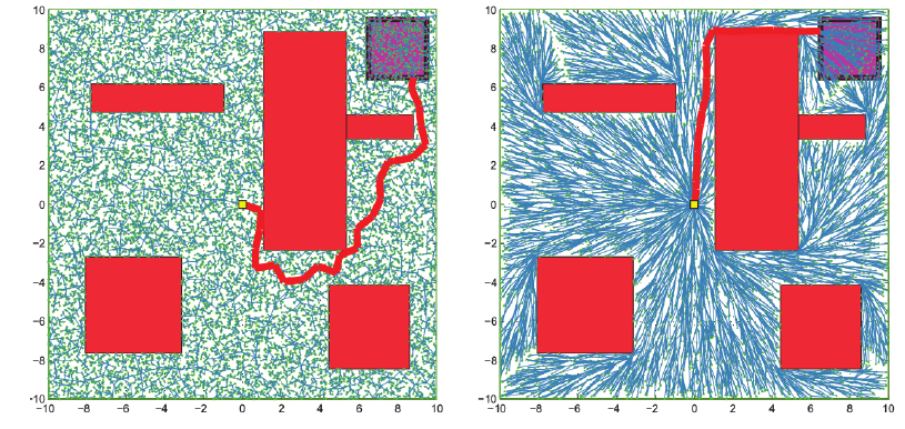
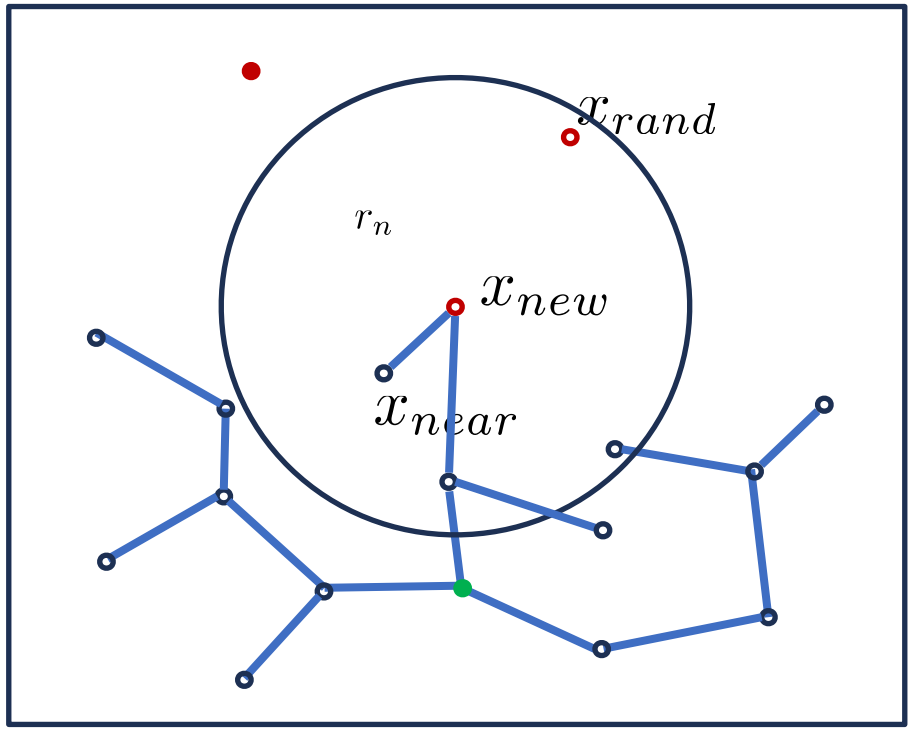




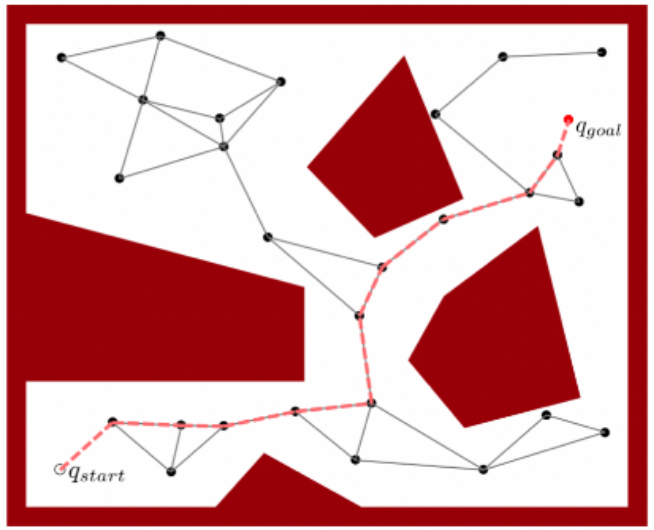
Observations

- PRM-like motion planning algorithms
  - For a given number of nodes n, they find “good” paths
  - ...however, require many costly collision checks

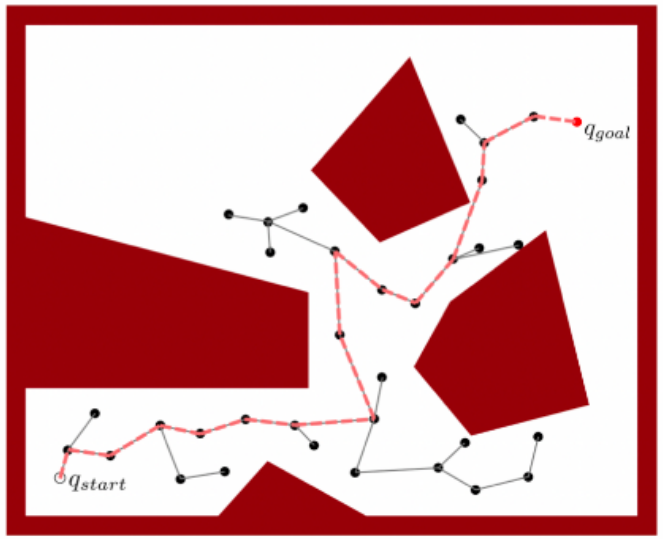
- RRT-like motion planning algorithms
  - Finds a feasible path quickly
  - ...however, the quality of that path is, in general, poor
  - “traps” itself by disallowing new better paths to emerge - RRT* performs local label correction as samples are added to help remedy this

Which of these two roadmaps has been created by RRT?

(A)



(B)



(A)

(B)

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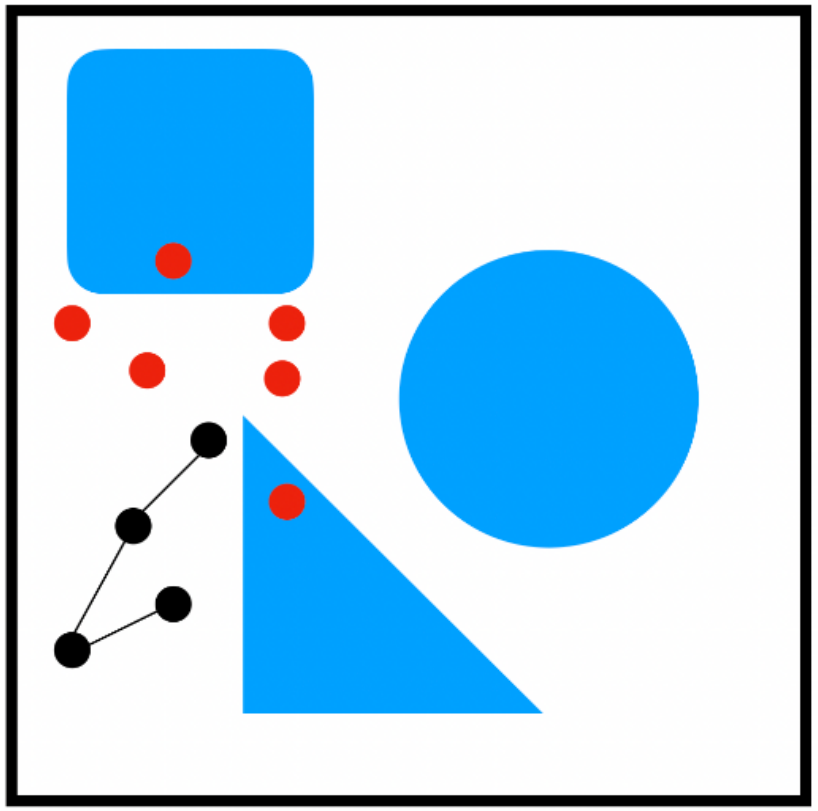
How many of the red sample points could lead to a new node in the roadmap constructed by RRT?



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Details and notes:

- Sampling-based planners transform the difficult global problem into a large set of local and easy problems
- A key ingredient is collision checking, which can be computationally time consuming. Speed ups avoid collision checking
- Local planning (edge validation) is typically performed by dense sampling of path and collision detection
- Another key ingredient is nearest-neighbor search (best done with k-d tree): given a query point find its nearest neighbor(s) within a set of points -- also well studied theoretically and practically

Sampling-based planning

Pros:

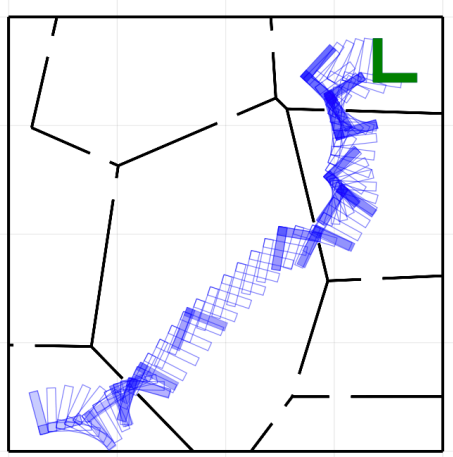
- Conceptually simple
- Relatively-easy to implement
- Flexible: one algorithm applies to a variety of robots and problems
- Beyond the geometric case: can cope with complex differential constraints, uncertainty, etc.

Cons:

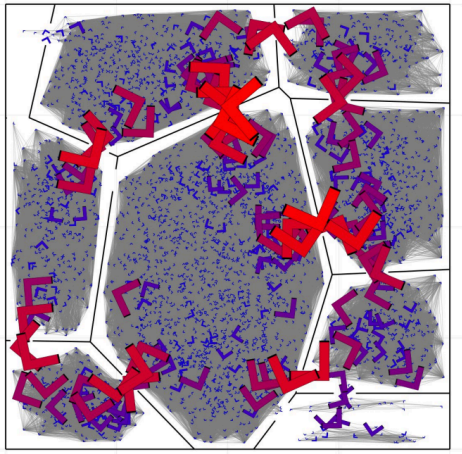
- Asymptotic
- Unclear how many samples should be generated to retrieve a solution
- Cannot determine whether a solution does not exist

Biased sampling for SBMP

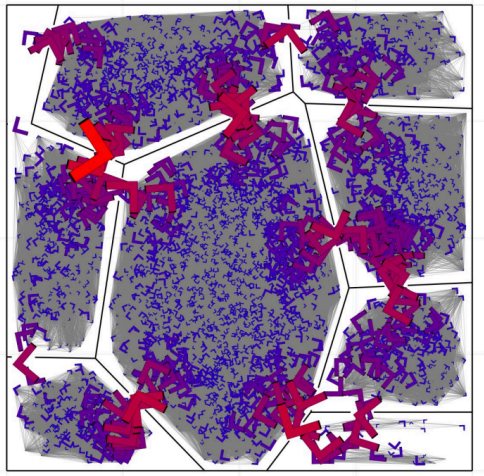
- Potential issue with uniform sampling: narrow corridors in C-space require many samples to identify/traverse
- Key idea: bias sampling towards suspected challenging regions of C-space
- Biased sampling distributions can be hand-constructed and/or adapt online (e.g., Hybrid Sampling PRM)
- Or learned from prior experience solving similar planning problems



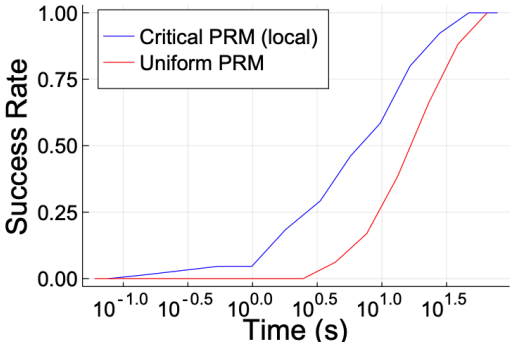
(a) SE(2) Planning



(b) Ground Truth Criticality



(c) Learned Criticality

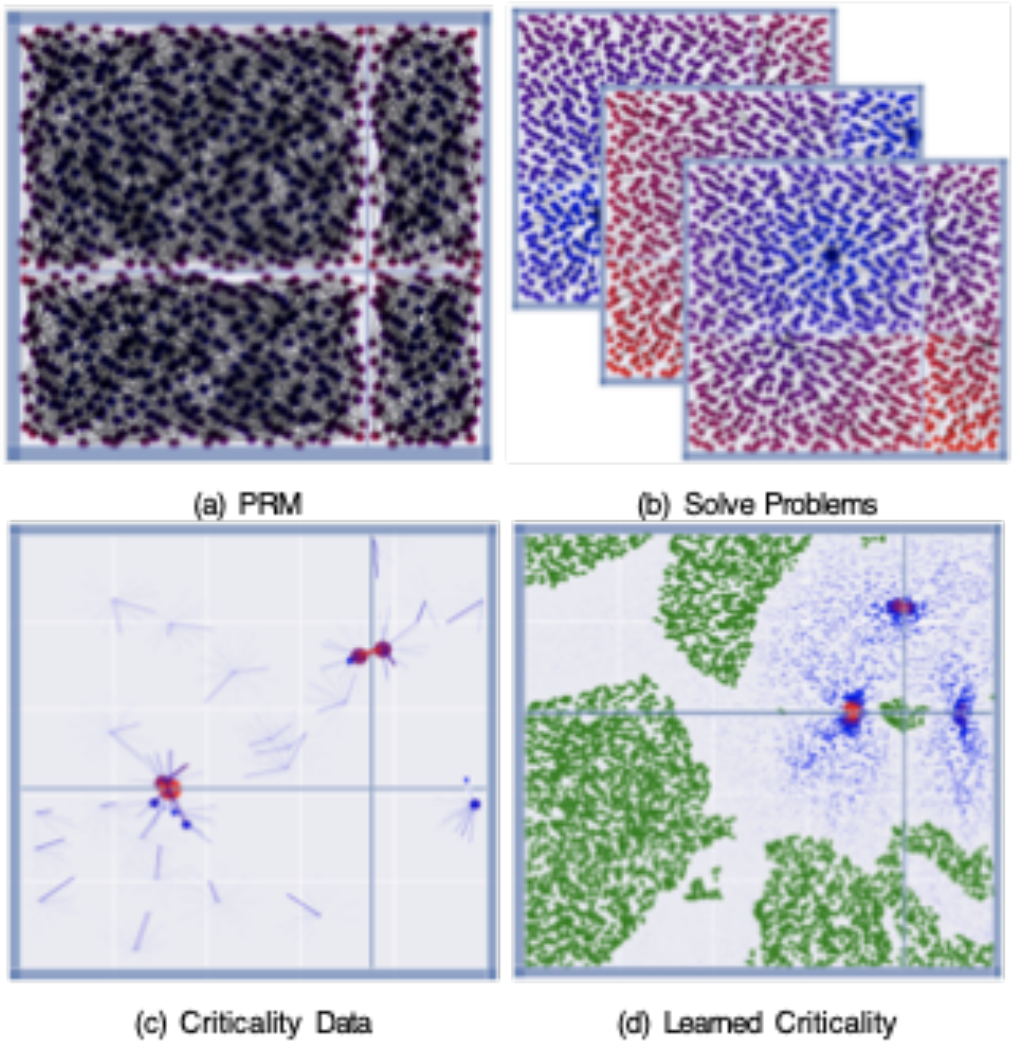


(d) Time (s) vs. Success

References:
Hsu et al. Hybrid PRM sampling with a cost-sensitive adaptive strategy. 2005.
Ichter et al. Learned Critical Probabilistic Roadmaps for Robotic Motion Planning. 2020

Learning criticality from local map features

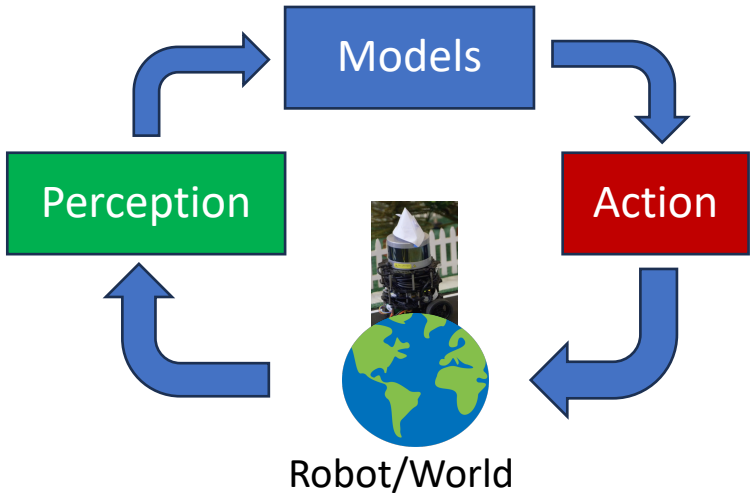
- On a collection of different maps, solve for optimal PRM paths between multiple start-goal pairs
- Score criticality of all nodes based on number of optimal paths passing through node
- Discount if node can be removed without collision
- Train CNN from local map patch centered at point to criticality of point



Ichter et al. Learned Critical Probabilistic Roadmaps for Robotic Motion Planning. 2020

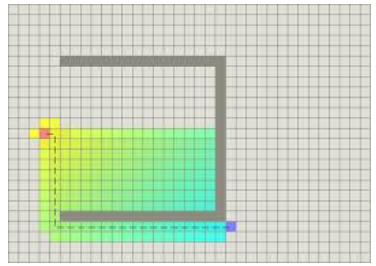
Action: Planning and Control Stack

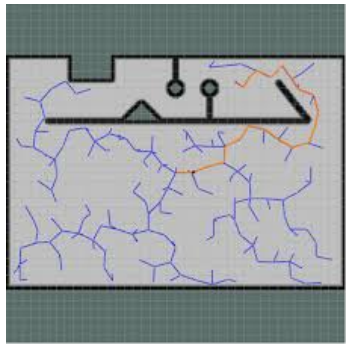
Use the models to move the robot



Path planning:

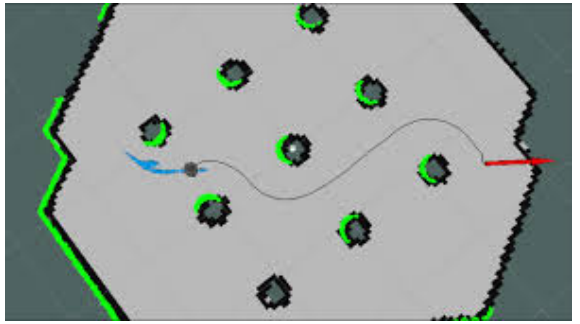
- A* planning on a graph
- RRT, RRT*





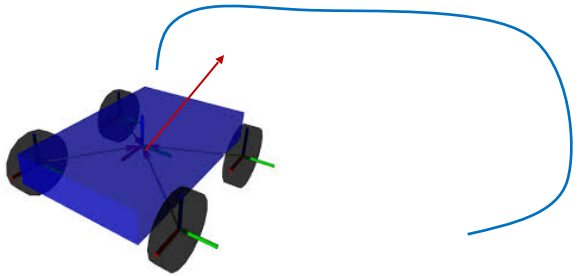
Trajectory optimization:

- Optimal control
- Numerical solvers



Trajectory following:

- Kinematic/dynamic models
- PID control
- LQR, gain scheduled LQR



Action: Planning and Control Stack

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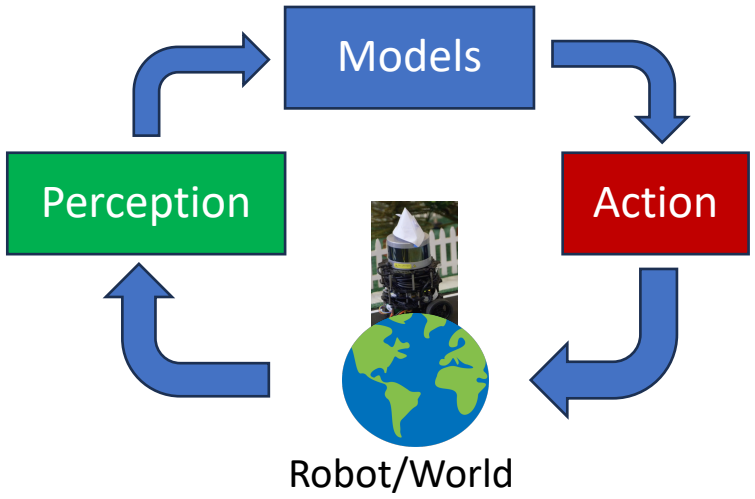

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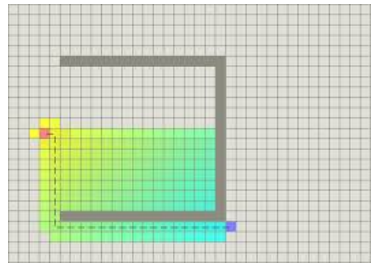

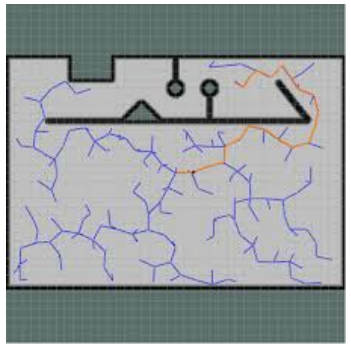

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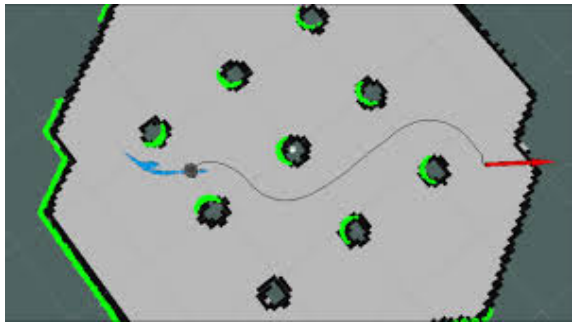

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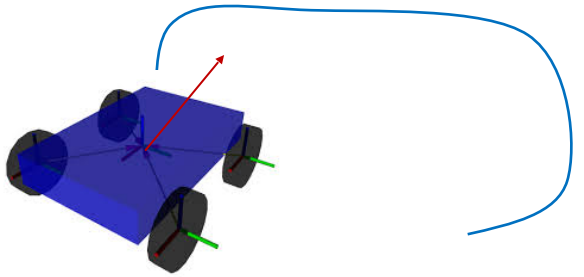
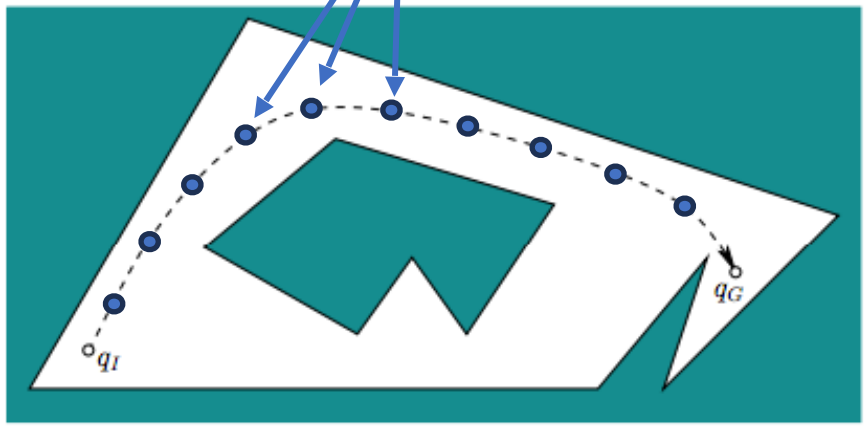


Trajectory Optimization

Waypoint poses from path planner
(e.g. A*, RRT*)



Trajectory optimization problem:

$$\min_{u_{0:T}} \sum_{t=0}^{T} c_t(x_t, u_t)$$

$$\text{s.t.} \quad x_{t+1} = f(x_t, u_t)$$

$$x_t \in \mathbb{X}_{\text{Free}}$$

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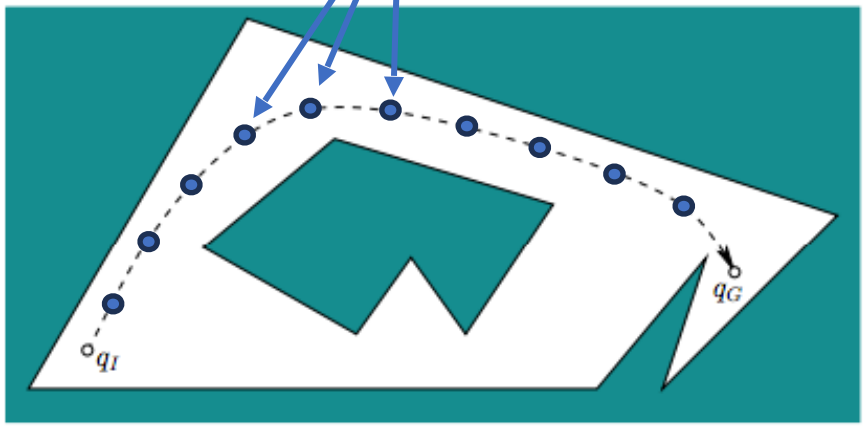

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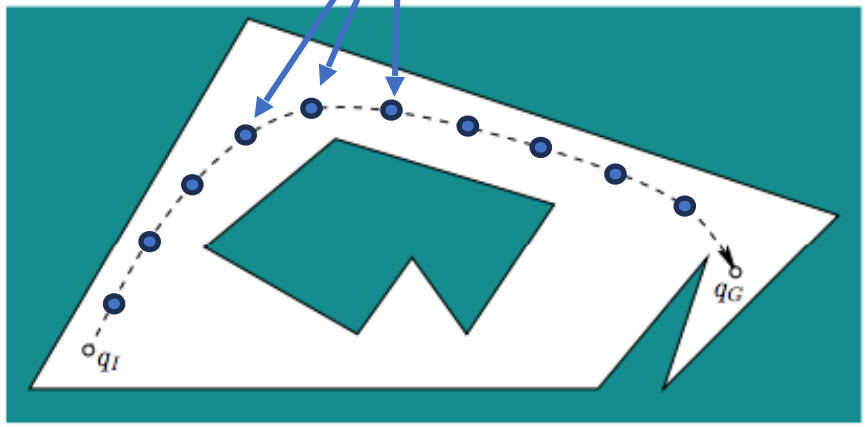

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$$x_t \in \mathbb{X}_{\text{Free}}$$

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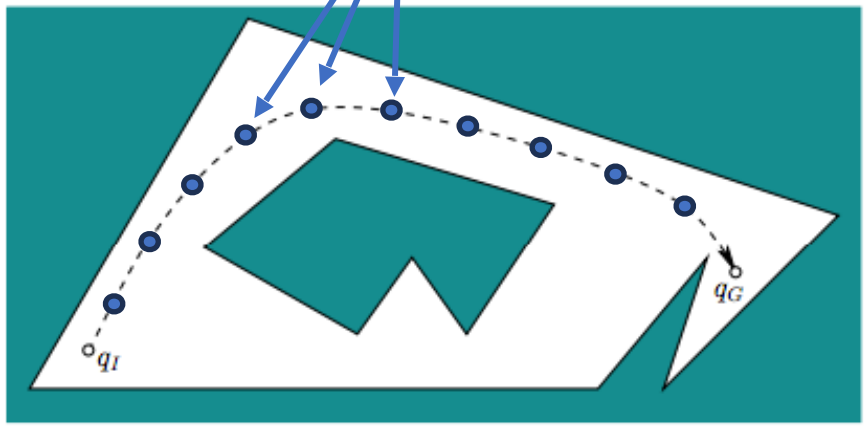

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$$\text{s.t.} \quad x_{t+1} = f(x_t, u_t)$$ Dynamic constraints

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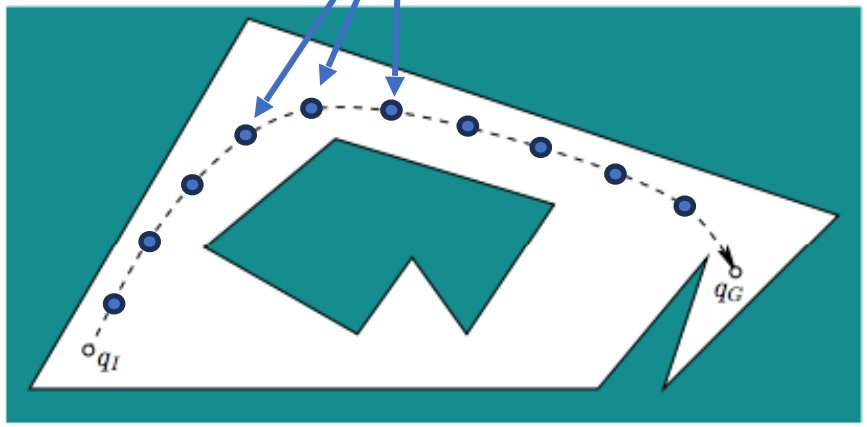
Trajectory Optimization

Waypoint poses from path planner
(e.g. A*, RRT*)



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$$x_t \in \mathbb{X}_{\text{Free}}$$ Collision constraints

Goal: Find a dynamically feasible trajectory (close to the path) so the robot doesn't hit things!